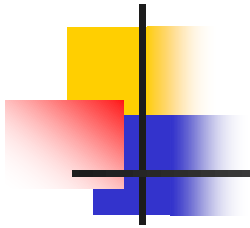
FUZZY SOFT COMPUTING METHODS AND THEIR APPLICATIONS IN ANALYTICAL CHEMISTRY

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Outline



- Introduction
- Classical Logic vs. Fuzzy Logic
- Crisp Sets vs. Fuzzy Sets
- Fuzzy Sets Theory and Objective Function
- Fuzzy Algorithms
- Applications
- Conclusions

Chemistry vs. Mathematics

"In any branch of study of the natural world, the amount of actual science contained therein is directly proportional to the amount of mathematics used. Chemistry can under no circumstances be regarded as a science"

Immanuel KANT **(1724 – 1804)**

Chemistry vs. Mathematics

"In physical science the first essential step in the direction of learning any subject is to find principles of numerical reckoning and practicable methods for measuring some quality connected with it. I often say that when you can measure what you are speaking about and express it in numbers, you know something about it; but when you cannot measure it, when you cannot express it in numbers, your knowledge is of a meager and unsatisfactory kind: it may be the beginning of knowledge but you have scarcely, in your thoughts, advanced to the state of science, whatever the matter may be"

***Lord KELVIN* (1824-1907)**

Chemistry vs. Mathematics

... all is number

("Lumea este guvernată de numere")

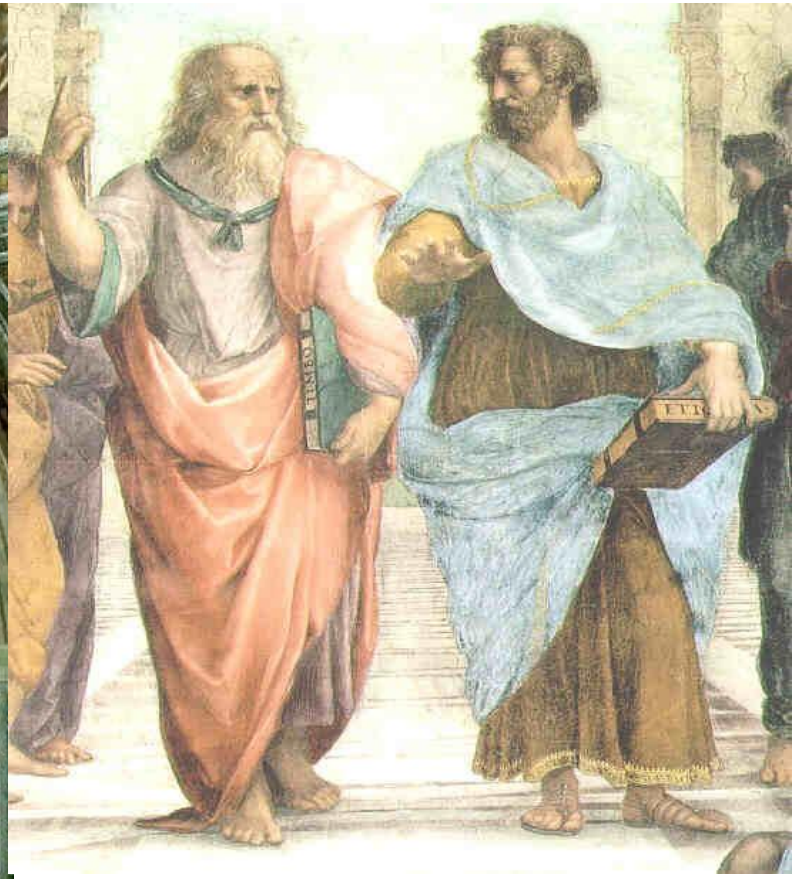
PITAGORA **(ca. 572-490 î.H.)**

"Când Dumnezeu a făcut lumea, a gândit-o în formule matematice"

LEIBNIZ **(1646-1716)**

School of Athens





School of Athens

- **Plato (428-328 BC)**
- The natural world we perceive through our senses (see, hear, touch etc.) reveals only a fallen, shadow, incomplete versions of this ideal truth.
- The true forms of natural things or of concepts exist in the way that mathematical truths or forms exist.
- Thus attempting to understand the truth of things by examining the natural world is not only folly, but likely dangerously misleading.
- As truth lies in the abstract and exists more clearly in our minds than in the natural world, philosophical contemplation, rather than observation, is the road toward truth.

- **Aristotle (384-322 BC)**
- Through the systematic observation and analysis (breaking down and classification) of the natural world, in combination with rigorous logic we can make true statements about the natural world and understand:
- (1) The nature of essences (i.e. what something is)
- (2) The nature of causes (i.e. why things occur)
- Argument (or dialectic) is a key ingredient for people reaching understandings: by arguing over issues the truth and falsity of the claims becomes increasingly apparent.

Chemistry vs. Statistics

Sample	Polution from air (g)	Polution from chemical decomposition (g)
1	**2.31017**	**2.30143**
2	**2.30986**	**2.29890**
3	**2.31010**	**2.29816**
4	**2.31001**	**2.30182**
5	**2.31024**	**2.29869**
6	**2.31010**	**2.29940**
7	**2.31028**	**2.29849**
8		**2.29889**
Average	**2.31011**	**2.29947**
Standard deviation	**0.00014**	**0.00138**

Chemistry vs. Mathematics

"Those knowing the meaning of the formula

$$\beta = (X'X)^{-1}X'Y$$

are invited to consider the job posting:

we hire specialists … "

(*Quarterly Journal of Economics III 1976*)

Chemometrics

Chemometrics is a chemical discipline that uses mathematical, statistical and other methods employing formal logic

1. To design or select optimal measurement procedures and experiments
2. To provide maximum relevant chemical information by analyzing chemical data

Computational Chemistry

Computational chemistry is a branch of chemistry that uses computers to assist in solving chemical problems. It uses the results of theoretical chemistry, incorporated into efficient computer programs, to calculate the structures and properties of molecules and solids. While its results normally complement the information obtained by chemical experiments, it can in some cases predict hitherto unobserved chemical phenomena. It is widely used in the design of new drugs and materials.

What is Soft Computing ?

- **Soft Computing** is a collection of methodologies (working synergistically, not competitively) which, in one form or another, reflect its guiding principle:

- **Exploit** the **tolerance** for **imprecision**, **uncertainty**, **approximate reasoning** and **partial truth** to achieve **tractability**, **robustness**, and **close resemblance** *with* ***human like decision making***

- Provides **flexible information processing capability** for representation and evaluation of **various real life ambiguous and uncertain situations**. ⇓ **Real World Computing**

- It may be argued that it is ***soft computing rather than hard computing*** that should be viewed as the foundation for **Artificial Intelligence** (**AI**).

Soft Computing vs. Hard Computing

Hard computing, i.e., conventional computing, requires a precisely stated analytic model and often a lot of computation time.

Soft computing is tolerant of imprecision, uncertainty, partial truth, and approximation. In effect, the role model for soft computing is the human mind.

Hard computing is based on binary logic, crisp systems, numerical analysis and crisp software.

Soft computing is based on fuzzy logic, neural nets and probabilistic reasoning.

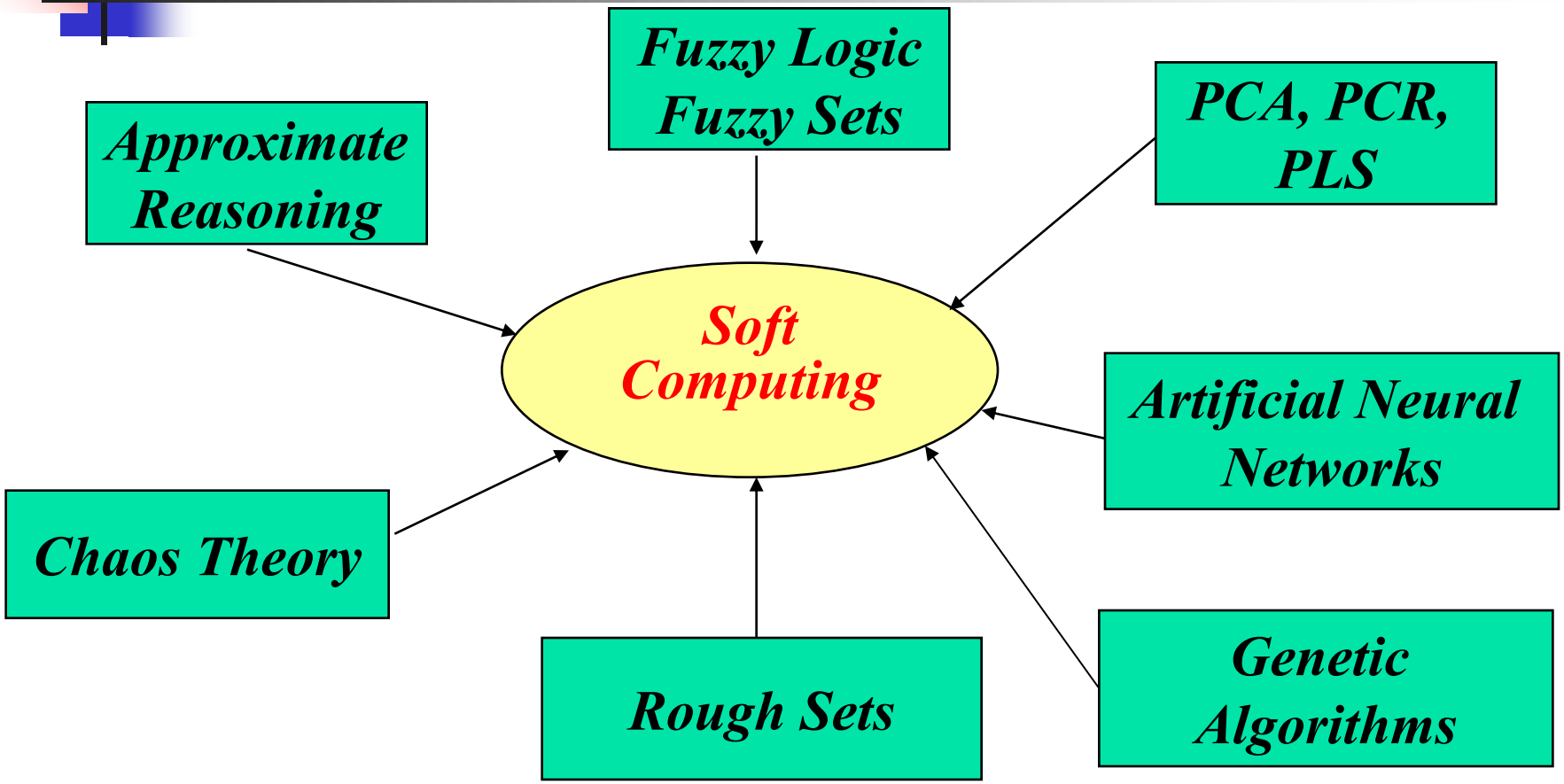
Hard computing has the characteristics of precision and categoricity. In hard computing, imprecision and uncertainty are undesirable properties.

In **soft computing** the tolerance for imprecision and uncertainty is exploited to achieve tractability, lower cost, high Machine Intelligence Quotient (MIQ) and economy of communication

Hard computing requires programs to be written, uses two-valued logic, is deterministic, requires exact input data, is strictly sequential, produces precise answers.

Soft computing can evolve its own programs, can use multi valued or fuzzy logic, incorporates stochastic, can deal with ambiguous and noisy data, allows parallel computations, can yield approximate answers.

Soft Computing Methods



Fuzzy Sets and **Fuzzy Logic**

- In 1965* **Zadeh** published his seminal work "**Fuzzy Sets**" which described the mathematics of **Fuzzy Set Theory**, and by extension **Fuzzy Logic**.
- It deals with the **uncertainty** and **fuzziness** arising from interrelated humanistic types of phenomena such **subjectivity**, **thinking**, **reasoning**, **cognition**, and **perception**. This type of uncertainty is characterized by structure that lack **sharp boundaries**. This approach provides a way to translate a linguistic model of the **human thinking process** into a mathematical framework for developing the computer algorithms for **computerized decision-making processes**.

*L. A. ZADEH, Fuzzy Sets, Information Control, **1965**, 8, 338-353.

Fuzzy Sets Theory

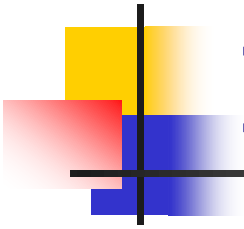
- A **Fuzzy Set** is a generalized set to which objects can belong with various **degrees** (grades) **of memberships** over the interval **[0,1]**.
- **Fuzzy systems** are processes that are too complex to be modeled by using **conventional mathematical methods**.
- In general, **fuzziness** describes objects or processes that are not amenable to **precise definition** or **precise measurement**. Thus, **fuzzy processes** can be defined as processes that are **vaguely defined** and have some **uncertainty in their description**. The data arising from **fuzzy systems** are in general, **soft**, with no precise boundaries.

Lotfi A. Zadeh
between Orient and Occident



Lotfi Zadeh during his student years in Tehran in the early 1940s (the large Russian sign ODIN, which means "alone," was his early proclamation of independence).

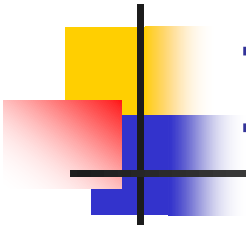




Is there a need for fuzzy logic?

"Is there a need for fuzzy logic?" is an issue which is associated with a long history of spirited discussions and debate. There are many misconceptions about fuzzy logic. Fuzzy logic is not fuzzy. Basically, fuzzy logic is a precise logic of imprecision and approximate reasoning. [...]"

Lotfi A. Zadeh, Is there a need for fuzzy logic?, *Information Sciences* 178, 13 (2008), 2751-2779



Is there a need for fuzzy logic?

"[...] More specifically, fuzzy logic may be viewed as an attempt at formalization/mechanization of two remarkable human capabilities. First, the capability to converse, reason and make rational decisions in an environment of imprecision, uncertainty, incompleteness of information, conflicting information, partiality of truth and partiality of possibility – in short, in an environment of imperfect information. And second, the capability to perform a wide variety of physical and mental tasks without any measurements and any computations."

Lotfi A. Zadeh, Is there a need for fuzzy logic?,
Information Sciences 178, 13 (2008), 2751-2779

The Impact of Application of Fuzzy Sets Theory in Science and Technical Fields

"In 1999, Japan exported products at a total of $35 billion that use Fuzzy Logic or Neuro-Fuzzy. The remarkable fact that an emerging key technology in Asia and Europe went unnoticed by the U.S. public until recently, combined with its unusual name and revolutionary concept has led to a controversial discussion among engineers."

Constantine von Altrock

Inform Software Corp., Germany

Reasoning Styles in Far East and West

Far East	West
Principle of Change Reality is a dynamical, constantly-changing process. The concepts that reflect reality must be subjective, active, flexible.	**Law of Identity** Everything is what it is. Thus it is a necessary fact that A equals A, no matter what A is.
Principle of Contradiction Reality is full of contradictions and never clear-cut or precise. Opposites coexist in harmony with one another, opposed but connected	**Law of Non-Contradiction** No statement can be both true and false.
Principle of Relationship To know something completely, it is necessary to know its relations, what it affects and what affects it.	**Law of the Excluded Middle** Every statement is either true or false. There is no middle term.

Fuzziness in Everyday World

- John is tall
- Temperature is hot
- Peter is bald
- That dress is red
- The Romanian Leu is getting relatively weak
- The people living close to Bucharest
- My car is slow, your car is fast

Fuzziness in Chemistry

- Water is an acid
- Germanium is a metal
- Those drugs are very effective
- Varying peaks in chromatograms
- Varying signal heights in spectra from the same substance
- Varying patterns in QSAR pattern recognition studies

Fuzziness in Nature



Fuzziness in Everyday World (Orient *vs.* Occident)





Characteristic Function in the Case of Crisp Sets and Fuzzy Sets

$P: X \rightarrow \{0,1\}$

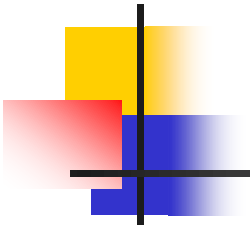
$P_{(x)} = 1$ if $x \in X$

$P_{(x)} = 0$ if $x \notin X$

$A : X \rightarrow [0,1]$

$A = \{x, A_{(x)}\}$ if $x \in X$

Fuzzy Approaches



- Fuzzy divisive hierarchical clustering
- Fuzzy horizontal clustering
- Fuzzy cross-clustering
- Fuzzy robust regression
- Fuzzy robust estimation of mean and spread

Generalized Fuzzy c-Means Algorithm

$$J(P,L)=\sum_{i=1}^{c}\sum_{j=1}^{n}(A_i(x^j))^m\, d^2(x^j,L^i)$$

$$A_i(x^j)=\frac{C(x^j)}{\sum_{k=1}^{c}\left(\frac{d^2(x^j,L^i)}{d^2(x^j,L^k)}\right)^{1/m-1}};\qquad L^i=\frac{\sum_{j=1}^{n}(A_i(x^j))^m\, x^j}{\sum_{j=1}^{n}(A_i(x^j))^m}$$

$$i=1,\ldots,c;\qquad j=1,\ldots,n$$

Fuzzy 1-Line Regression Algorithm

$$J(P,L,\alpha)=\sum_{j=1}^{n}(A(x^j))^m d^2(x^j,L)+\sum_{j=1}^{n}(\overline{A}(x^j))^m.\frac{\alpha}{1-\alpha}$$

$$A(x^j)=\frac{\left(\frac{\alpha}{1-\alpha}\right)^{1/m-1}}{\left(\frac{\alpha}{1-\alpha}\right)^{1/m-1}+\left(d^2(x^j,L)\right)^{1/m-1}};\qquad L=\frac{\sum_{j=1}^{n}(A(x^j))^m x^j}{\sum_{j=1}^{n}(A(x^j))^m}$$

$$j=1,\ldots,n$$

Fuzzy Principal Component Analysis Algorithm

1. Determine the best value of α. For this, loop with α between 0 and 1. For each iterative value of α minimize the objective function above, and, with the optimal membership degrees $A(x^j)$, compute the largest eigenvalue of the matrix C given below. Select the optimal value of α according to the maximal eigenvalue.

$$C_{kl} = \frac{\sum_{j=1}^{n} A(x^j)^m (x_{jk} - \bar{x}_k)(x_{jl} - \bar{x}_l)}{\sum_{j=1}^{n} A(x^j)^m}$$

Relevant Data Sets Concerning Robust Methods Comparison Studies

- *Case 1*: 9.08, 9.29, 9.32, 10.57, 10.72, 11.05, 10.36, 10.78, 10.33, 11.42, 10.43, 9.82, 10.41, 10.93, 9.40, 8.94, 8.35, 9.14, 9.92, 8.66, 8.45, 10.35, 10.19, 9.47, 9.24, 11.55, 11.12, 10.16, 9.43, 9.87, 10.63, 10.28, 11.34, 9.07, 9.43, 9.39, 9.68, 11.98, 9.90, 10.08, 10.04, 9.67, 9.13, 8.92, 9.50, 9.71, 10.74, 11.13, 9.82, 11.09

- *Case 2*: 9.08, 9.29, 9.32, 10.57, 10.72, 11.05, 10.36, 10.78, 10.33, 11.42, 10.43, 9.82, 10.41, 10.93, 9.40, 8.94, 8.35, 9.14, 9.92, 8.66, 8.45, 10.35, 10.19, 9.47, 9.24, 11.55, 11.12, 10.16, 9.43, 9.87, 10.63, 10.28, 11.34, 9.07, 9.43, 9.39, 9.68, 11.98, 9.90, 10.08, 10.04, ***15.67, 15.13, 14.92, 15.50, 15.71, 16.74, 17.13, 15.82, 17.09***

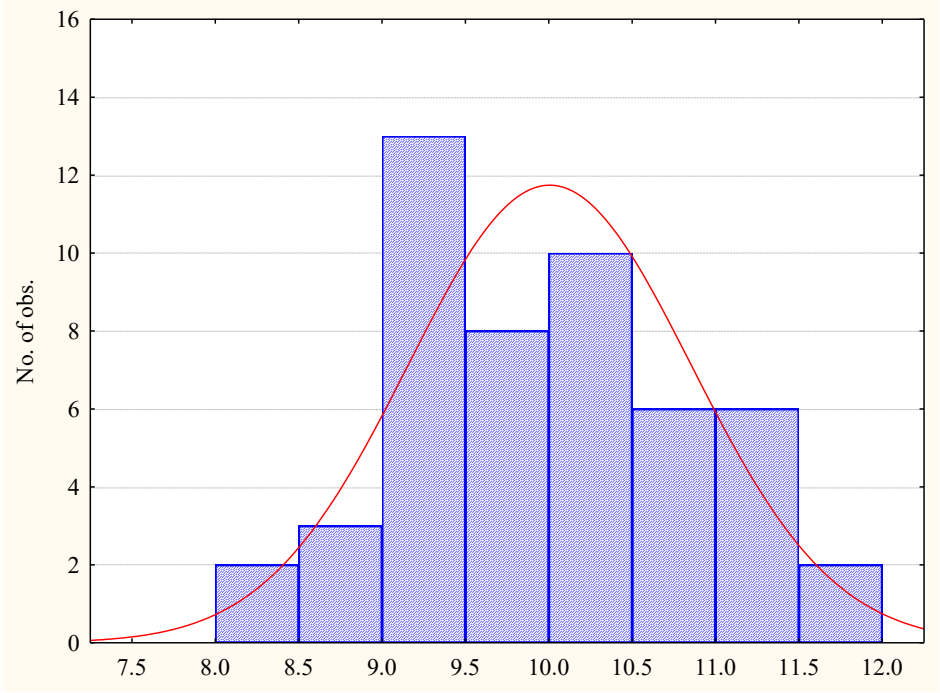
- *Case 3*: 9.08, 9.29, 9.32, 10.57, 10.72, 11.05, 10.36, 10.78, 10.33, 11.42, 10.43, 9.82, 10.41, 10.93, 9.40, 8.94, 8.35, 9.14, 9.92, 8.66, 8.45, 10.35, 10.19, 9.47, 9.24, 11.55, 11.12, 10.16, 9.43, 9.87, 10.63, 10.28, 11.34, 9.07, 9.43, 9.39, ***15.68, 17.98, 15.90, 16.08, 16.04, 15.67, 15.13, 14.92, 15.50, 15.71, 16.74, 17.13, 15.82, 17.09***

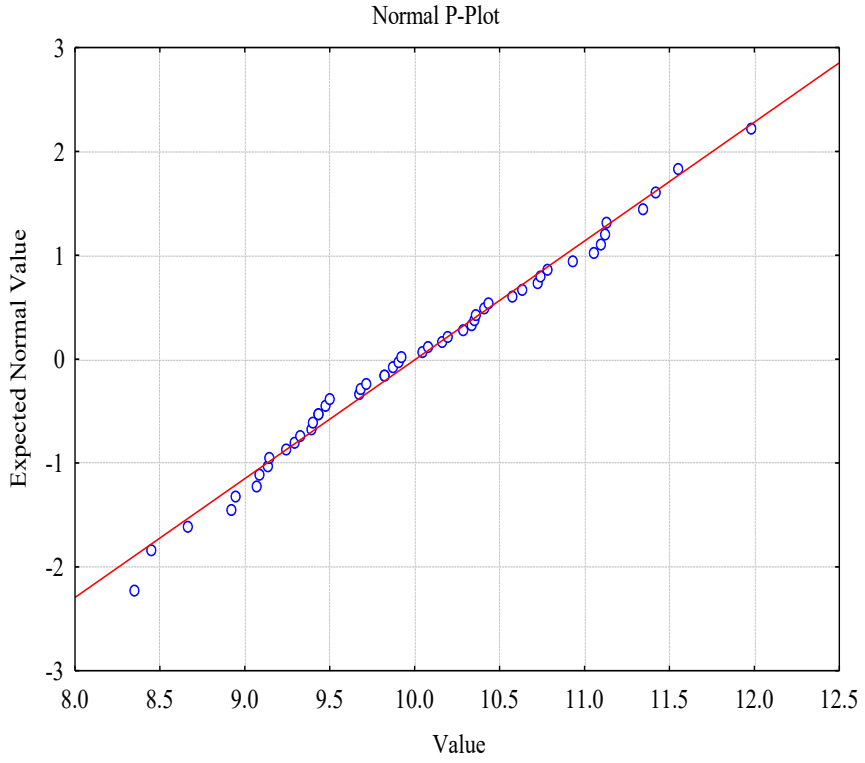
Robust Methods for Estimation of Central Location

Statistical parameter	Case 1	Case 2	Case 3
Mean	**10.006**	**11.086**	**11.686**
Median	**9.910**	**10.305**	**10.420**
5% Trimmed mean	**9.998**	**10.901**	**11.545**
Huber (1.339)	**9.950**	**10.314**	**10.585**
Hampel (1.7,3.4, 8.5)	**9.980**	**10.149**	**10.418**
Tukey (4.685)	**9.948**	**10.008**	**9.969**
Andrews (1.43)	**9.948**	**10.009**	**9.975**
Fuzzy mean	**9.828**	**10.035**	**10.074**

Fuzzy Mean is the Most Suitable (Best) Estimator of Central Location?

SD = 0.8493; Skewness = 0.2084; Kurtosis = -0.5661





Estimation Methods of Spread

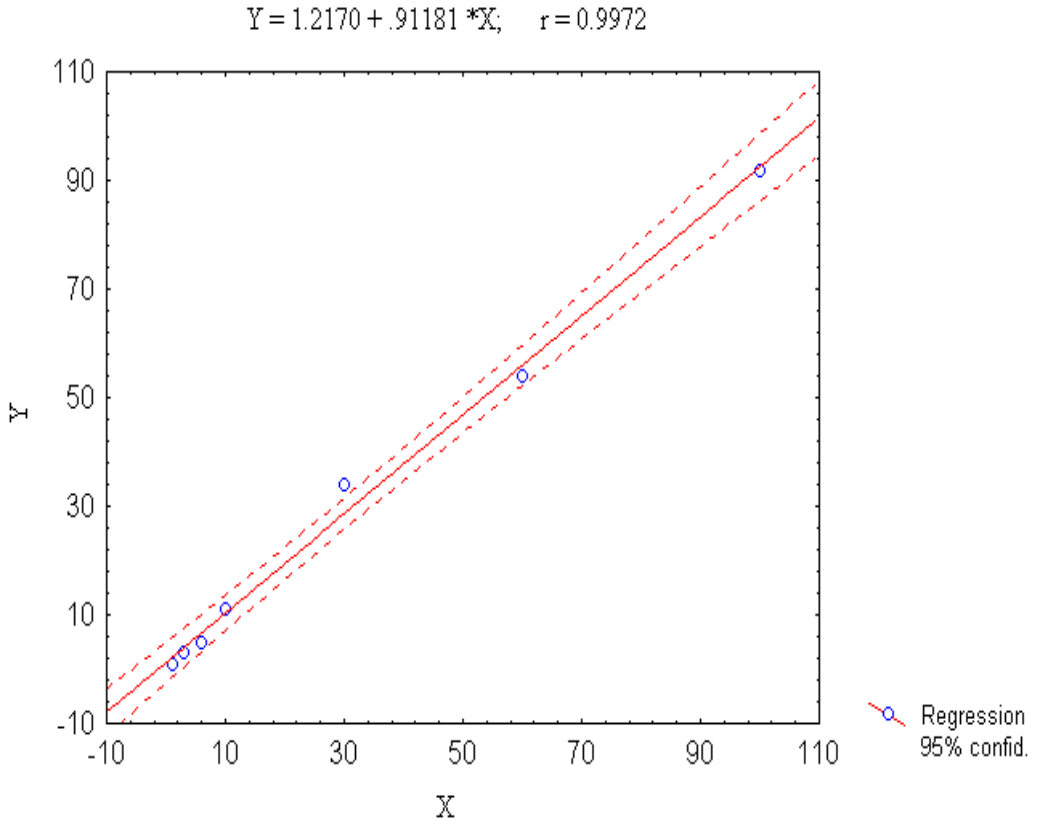
Statistical parameter	Relation	Case 1	Case 2	Case 3
Standard deviation	$SD = (\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2)^{1/2}$	**0.893**	**2.503**	**2.964**
Standard error	$s_e = SD/n^{1/2}$	**0.120**	**0.354**	**0.419**
Range	$R = x_{max} - x_{min}$	**3.630**	**8.780**	**9.630**
Interquartile range	$IQR = Q_3 - Q_1$	**1.280**	**1.937**	**5.987**
Informational energy	$E = \frac{1}{N}\sum_{j=1}^{N}\left(A(x^j)\right)^2$	**0.675**	**0.621**	**0.563**
Informational entropy	$H = -\frac{1}{N}\sum_{j=1}^{N}A(x^j)\log_2 A(x^j)$	**0.151**	**0.137**	**0.149**

Fuzzy Regression: Heteroscedastic Data

Case 1. (Garden, J.S. et al., *Anal. Chem.*, 1980, 52, 2310.)

X	Y
1	1
3	3
6	5
10	11
30	34
60	54
100	92

$y = 0.150 + 0.950*x$; $w_i = s_i^{-2}$



Quality Coefficients of Goodness of Fit

$$QC_1 = \sqrt{\frac{\sum_{i=1}^{N}((y_i - \hat{y}_i)/\hat{y}_i)^2}{N-1}}$$

$$QC_3 = \sqrt{\frac{\sum_{i=1}^{N}((y_i - \hat{y}_i)/\bar{\hat{y}}_i)^2}{N-1}}$$

$$QC_5 = \sqrt{\sum_{i=1}^{N}(r_i / \max|r_i|)^2}$$

$$\mathrm{NQC}_5 = \frac{QC_5 - 1}{\sqrt{N} - 1}$$

$$QC_6 = \sqrt{\sum_{i=1}^{N}(r_i / \bar{r})^2}$$

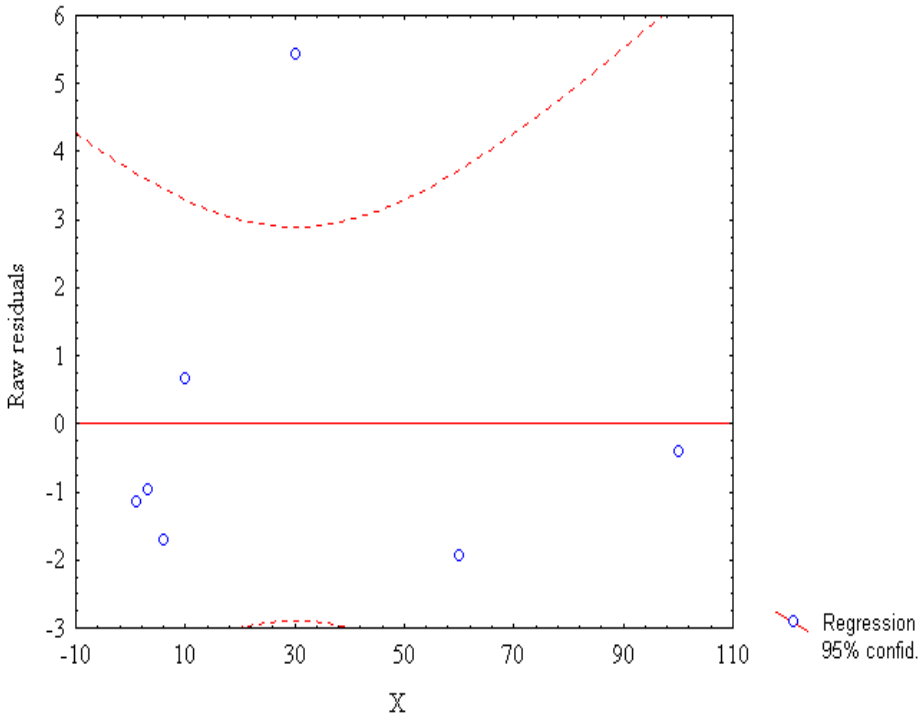
$$\mathrm{NQC}_6 = \frac{QC_6 - \sqrt{N}}{N - \sqrt{N}}$$

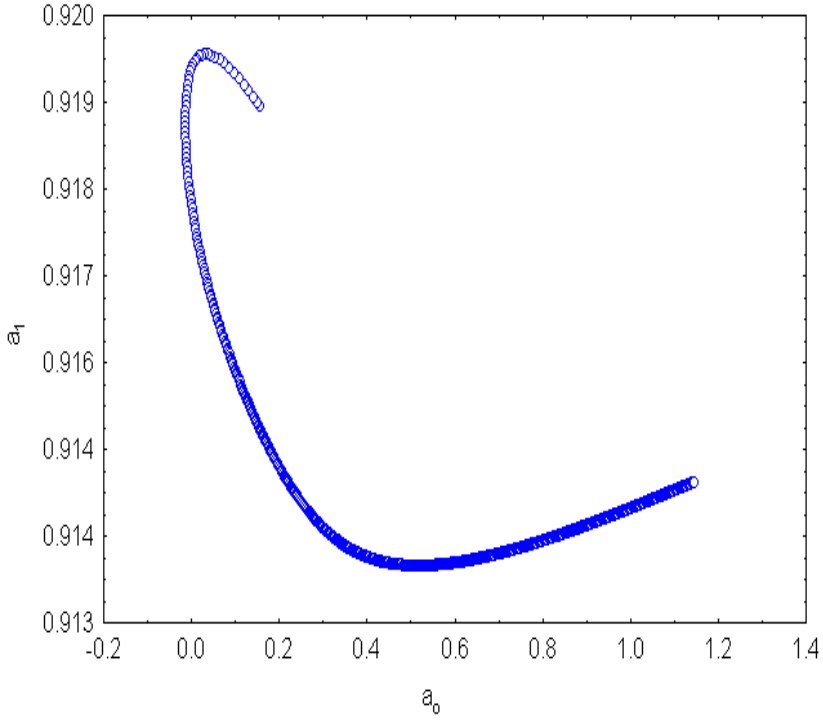
Comparative Regression Results. Case 1

($^1w_i = s_i^{-2}$; $^2w_i = x_i^{-2}$; $^3\alpha = 0.01$)

Method	a_1	a_o	QC_1	QC_5	NQC_5	QC_6	NQC_6
OLS	0.912	1.22	1.15	1.15	0.09	3.56	0.21
WLS[1]	0.950	0.15	1.33	1.33	0.20	3.57	0.21
XWLS[2]	0.980	0.02	1.50	1.50	0.30	3.63	0.22
FR[3]	0.919	0.03	1.35	1.06	0.39	4.63	0.45

Fuzzy Pattern Recognition of Heteroscedasticity in Comparing to Residuals Plot (Case 1)



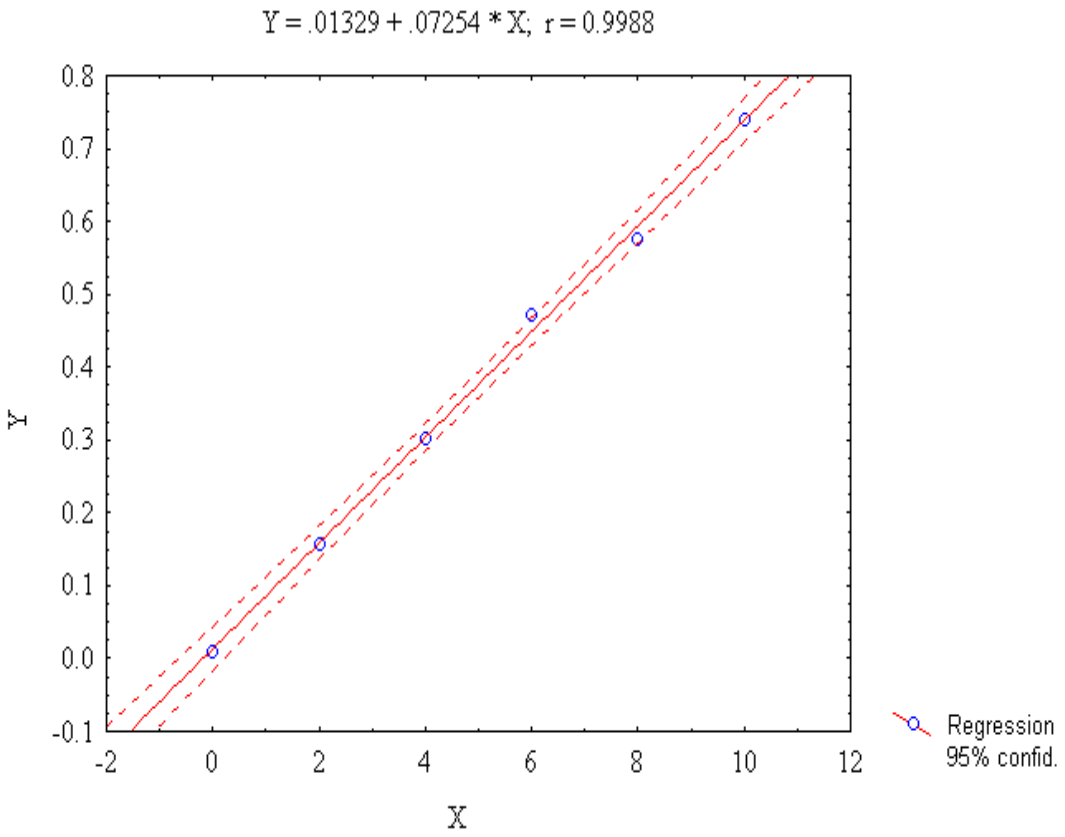


Fuzzy Regression: Heteroscedastic Data

Case 2. (Miller, J.C. and Miller J.N., *Statistics for Analytical Chemistry*, 2nd ed., Ellis Horwood, Chichester, 1988, p. 101)

$y = 0.009 + 0.074*x$; $w_i = s_i^{-2} x \Sigma\, s_i^{-2}/N$

X	Y
0	0.009
2	0.158
4	0.301
6	0.472
8	0.577
10	0.739

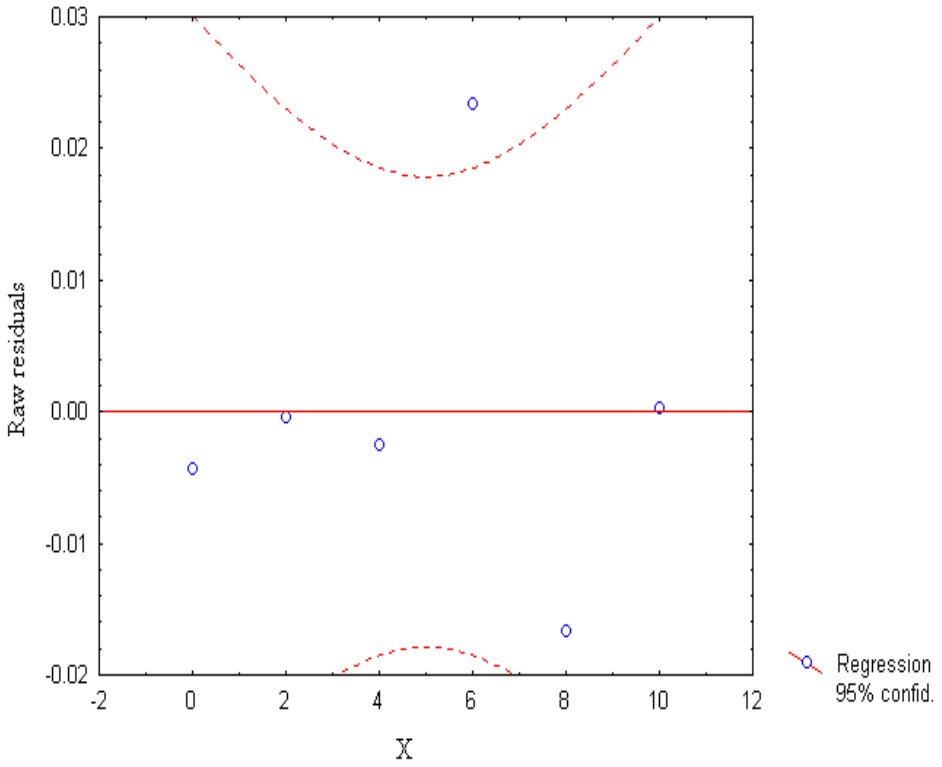


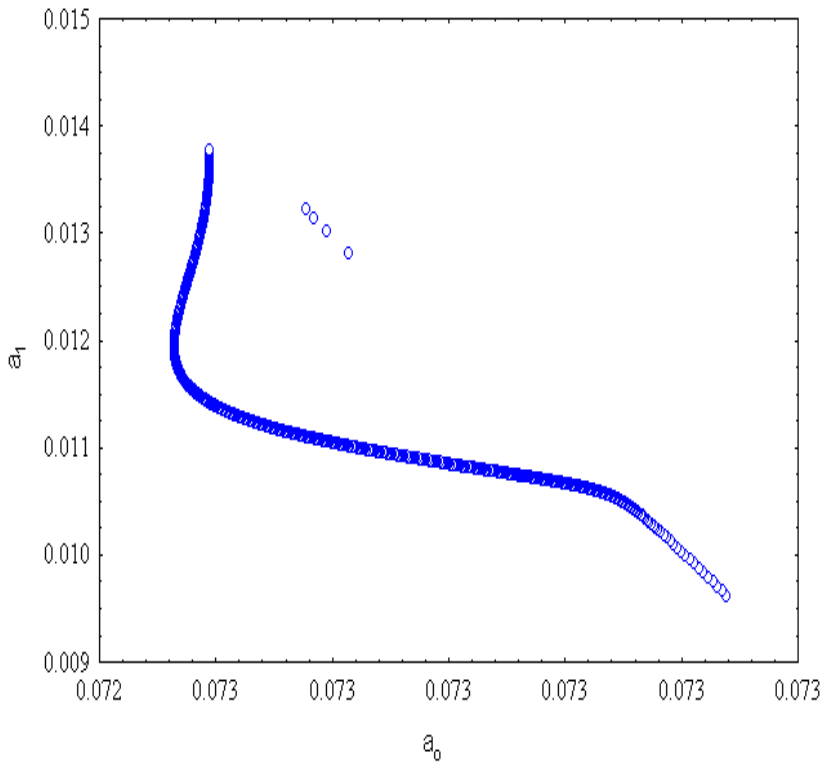
Comparative Regression Results. Case 2

($^1w_i = s_i^{-2} x \Sigma\, s_i^{-2}/N$; $^2w_i = x_i^{-2}$; $^3\alpha = 0.01$)

Method	a_1	a_o	QC_5	NQC_5	QC_6	NQC_6
OLS	0.072	0.013	1.22	0.15	3.69	0.35
WLS[1]	0.074	0.009	1.39	0.27	3.40	0.26
XWLS[2]	0.074	0.001	1.43	0.29	3.45	0.28
FR[3]	0.073	0.009	1.19	0.13	3.97	0.43

Fuzzy Pattern Recognition of Heteroscedasticity in Comparing to Residuals Plot (Case 2)



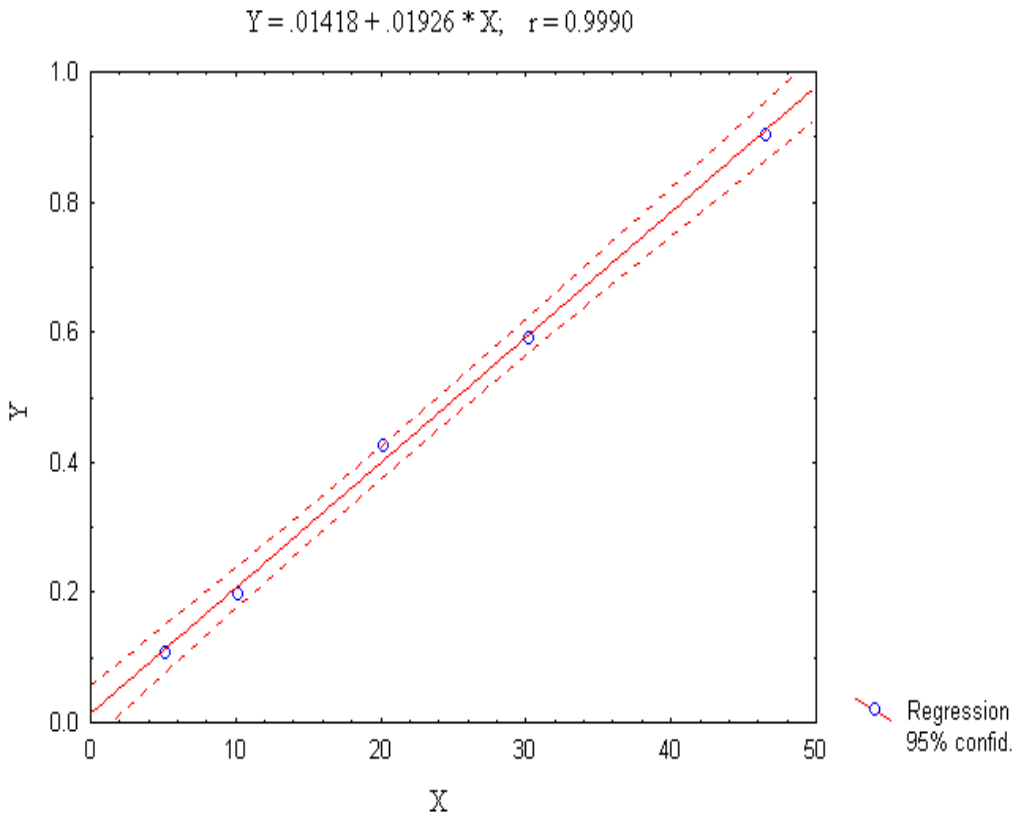


Fuzzy Regression: Heteroscedastic Data

Case 3. (Aarons, L., *J. Pharm. Biomed. Anal*, 1984, 2, 395.)

X	Y
5.1	0.1083
10.1	0.1977
20.2	0.4278
30.2	0.5916
46.5	0.9050

$y = 0.006 + 0.020*x;\ \ w_i = s_i^{-2} x \Sigma\, s_i^{-2}/N$

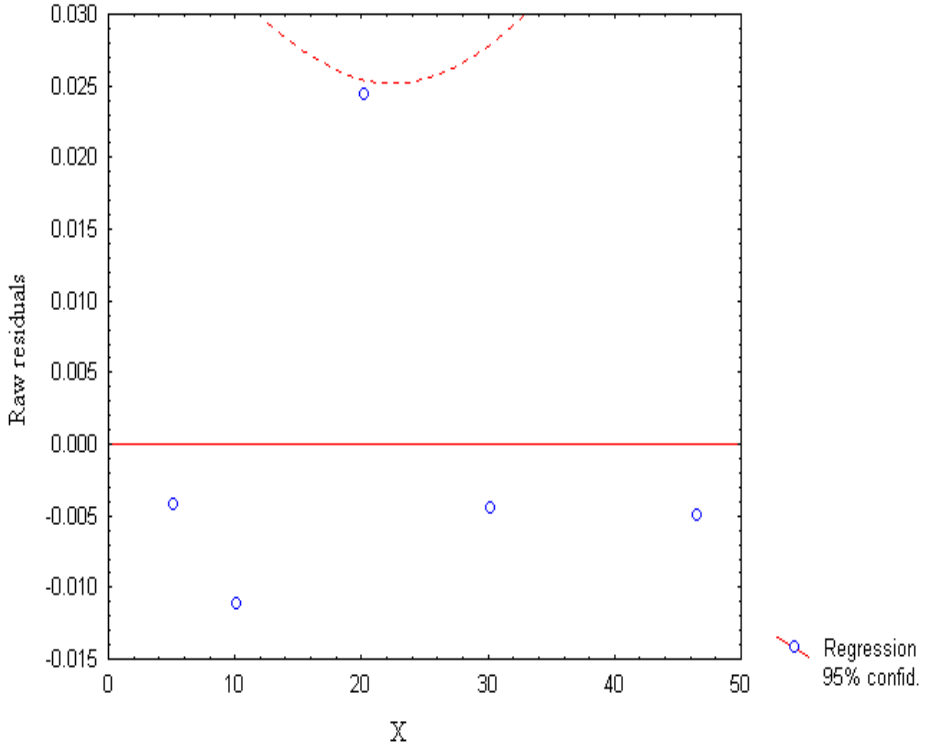


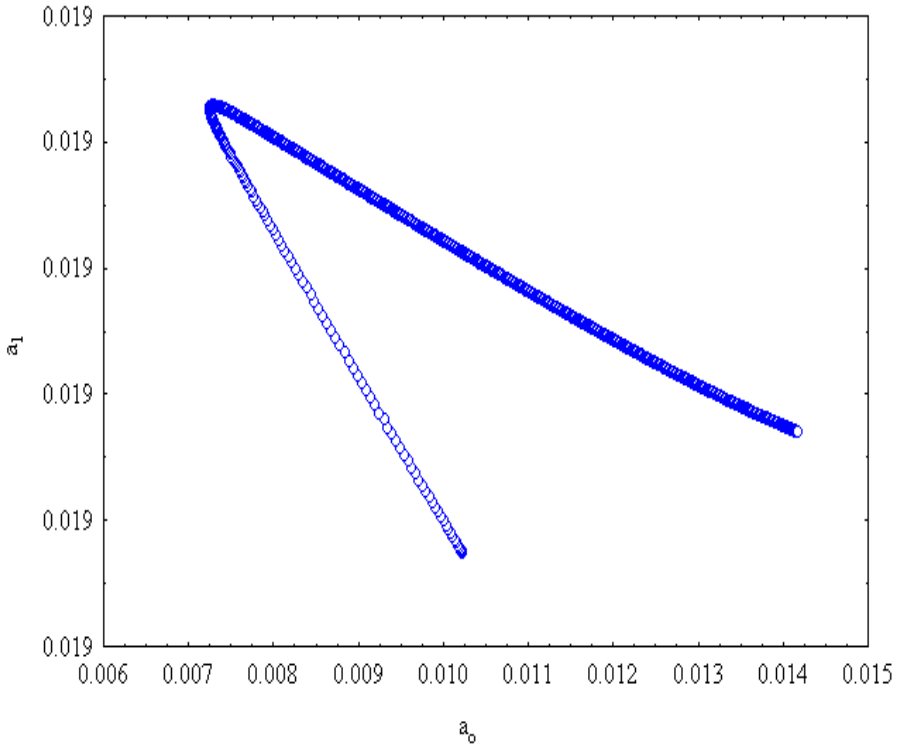
Comparative Regression Results. Case 3

($^1w_i = s_i^{-2} x \Sigma\, s_i^{-2}/N$; $^2w_i = x_i^{-2}$; $^3\alpha = 0.05$)

Method	a_1	a_o	QC_5	NQC_5	QC_6	NQC_6
OLS	0.019	0.014	1.17	0.14	2.73	0.17
WLS[1]	0.020	0.006	1.43	0.35	2.61	0.13
XWLS[2]	0.020	0.007	1.22	0.18	2.76	0.19
FR[3]	0.020	0.08	1.01	0.01	4.00	0.64

Fuzzy Pattern Recognition of Heteroscedasticity in Comparing to Residuals Plot (Case 3)

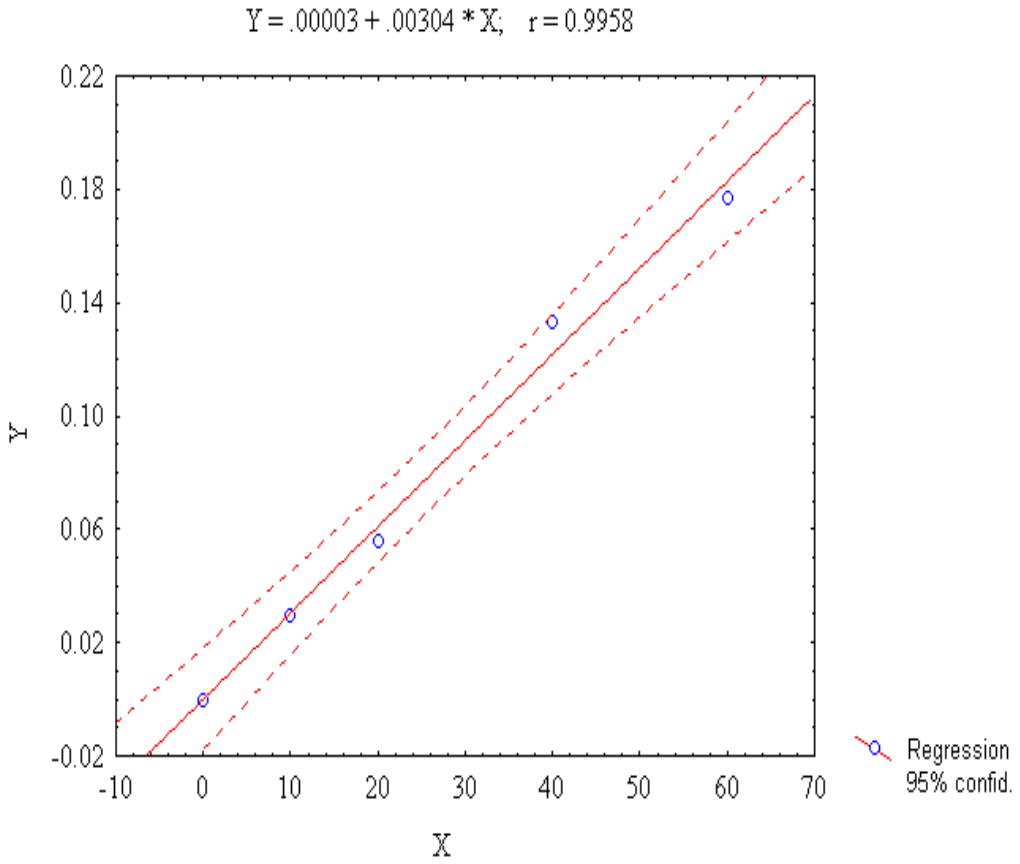




Fuzzy Regression: Data with Outliers.

1 Outlier. (Hu, Y. et al., *J. Anal. At. Spectrom.*, 1989, 4, 605.)

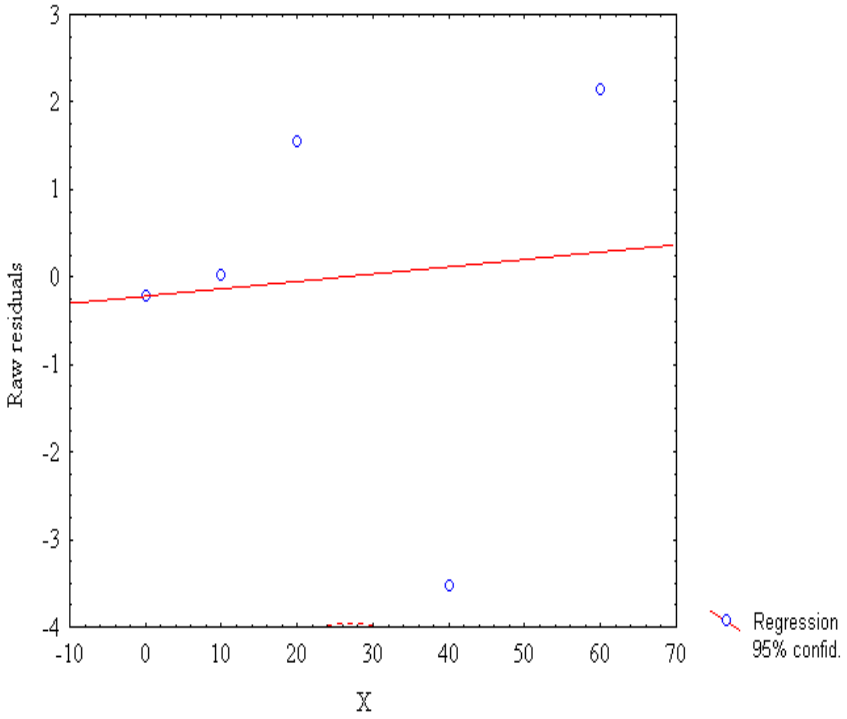
X	Y
0.0	0.000
10	0.030
20	0.056
40	0.133
60	0.177

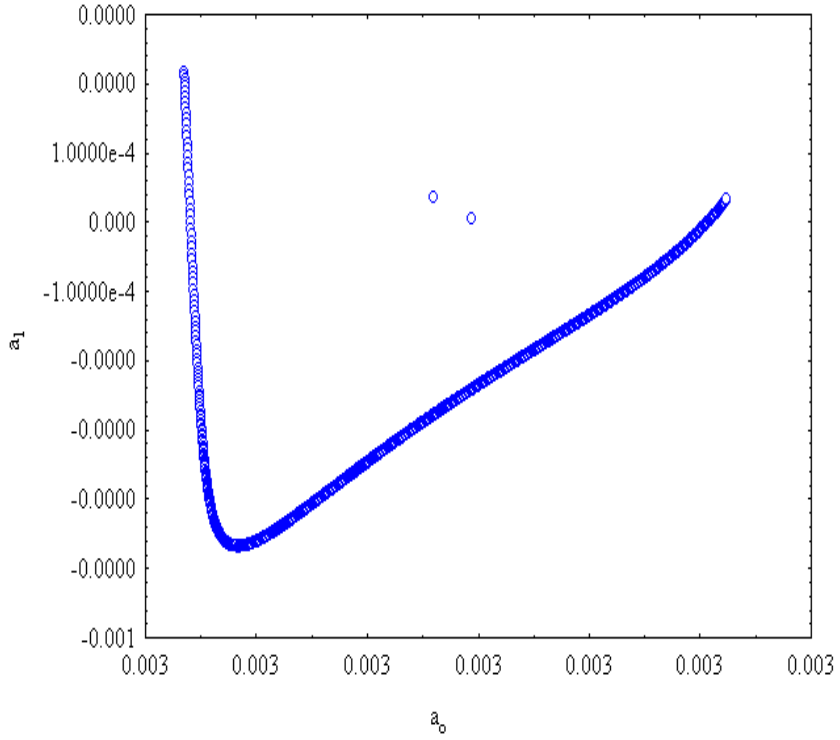


Comparative Regression Results. 1 Outlier ($^{*}x10^{-3}$)

Method	a_1^*	a_o^*	NQC_6
LS	3.04	0.03	0.29
LMS	2.95	0.00	0.68
FUZR	2.97	-1.11	0.58
FUZL	2.95	-0.04	0.68
FR	2.94	-0.15	0.68

Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)

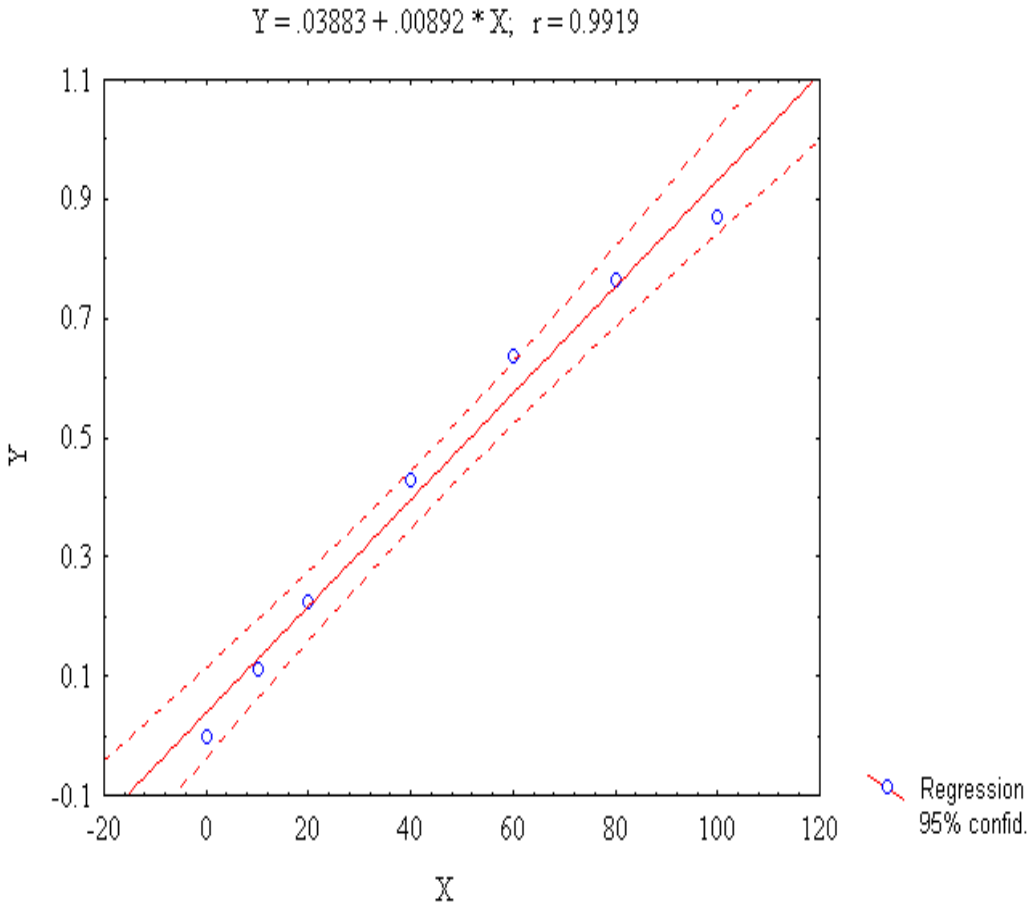




Fuzzy Regression: Data with Outliers.

2 Outliers. (Hu, Y. et al., *J. Anal. At. Spectrom.*, 1989, 4, 605.)

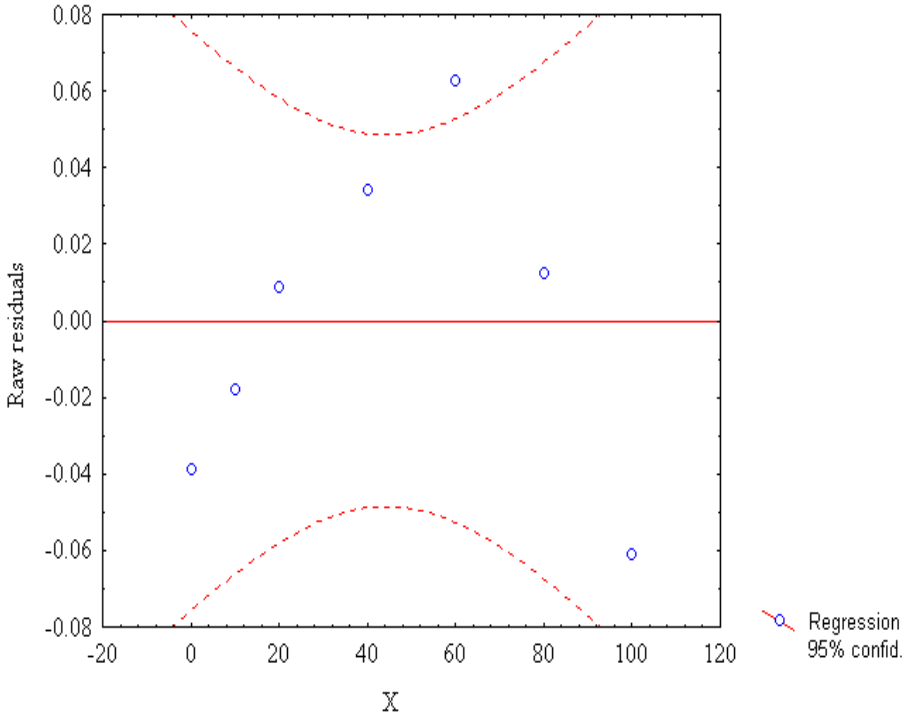
X	Y
0.0	0.000
10	0.110
20	0.226
40	0.430
60	0.637
80	0.765
100	0.870

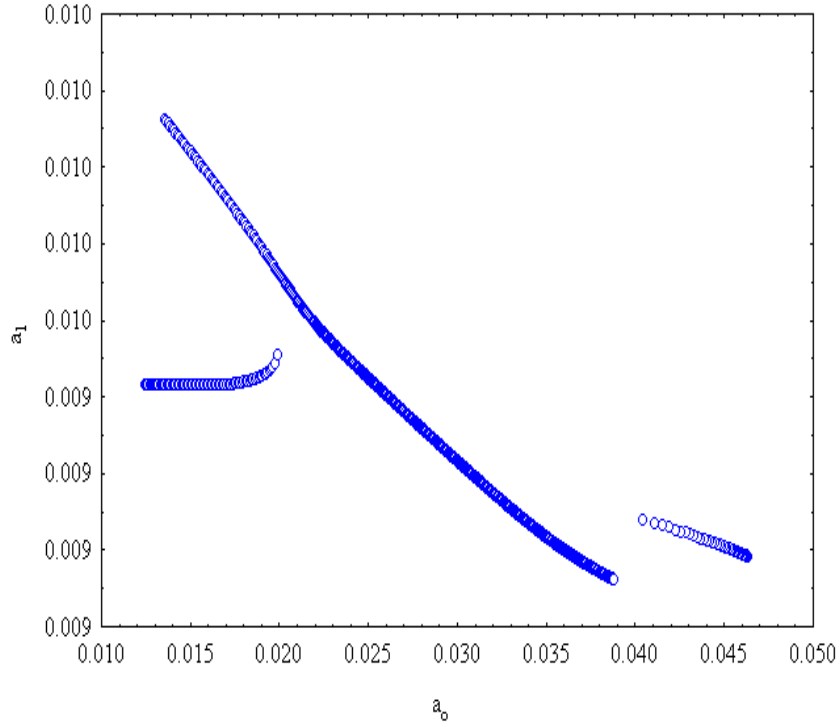


Comparative Regression Results. 2 Outliers

Method	a_1	a_o	NQC_6
LS	8.92	38.83	0.10
LMS	10.54	4.60	0.53
FUZR	10.40	8.74	0.49
FUZL	10.55	4.66	0.53
FR	9.44	12.11	0.20

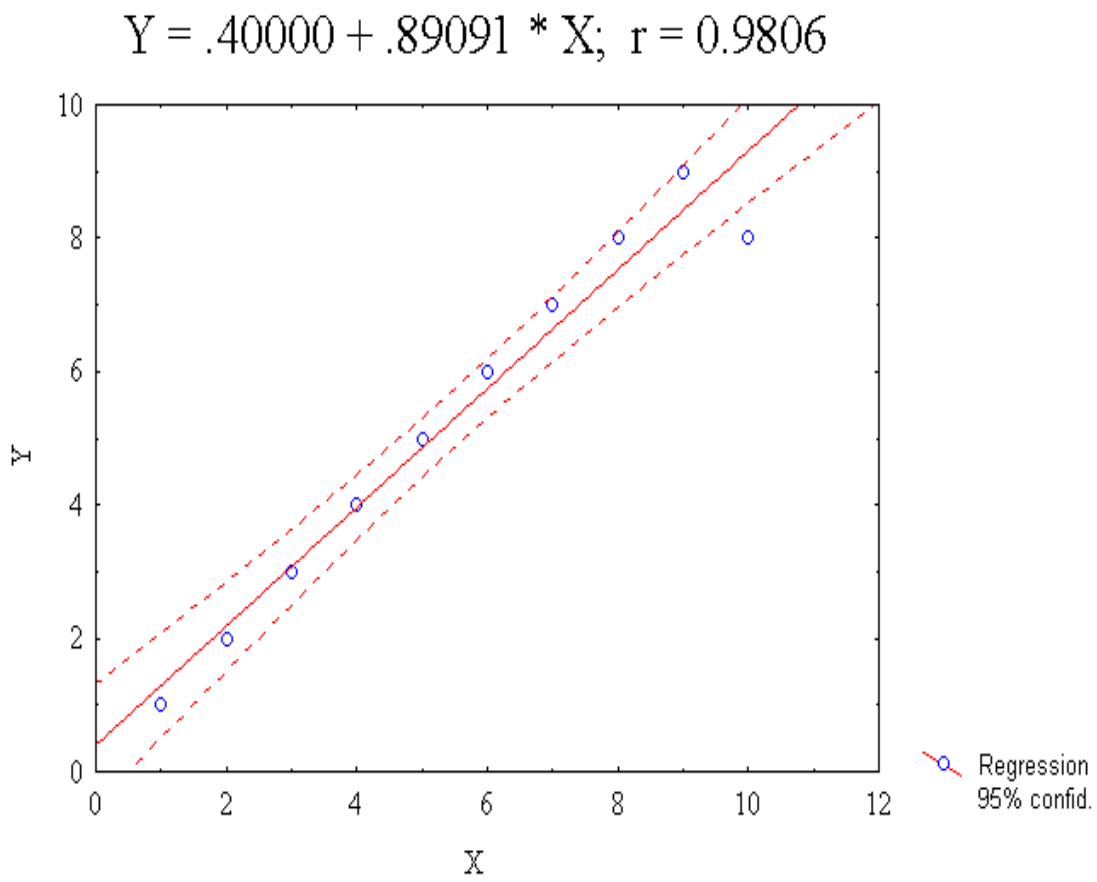
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (2 Outliers)



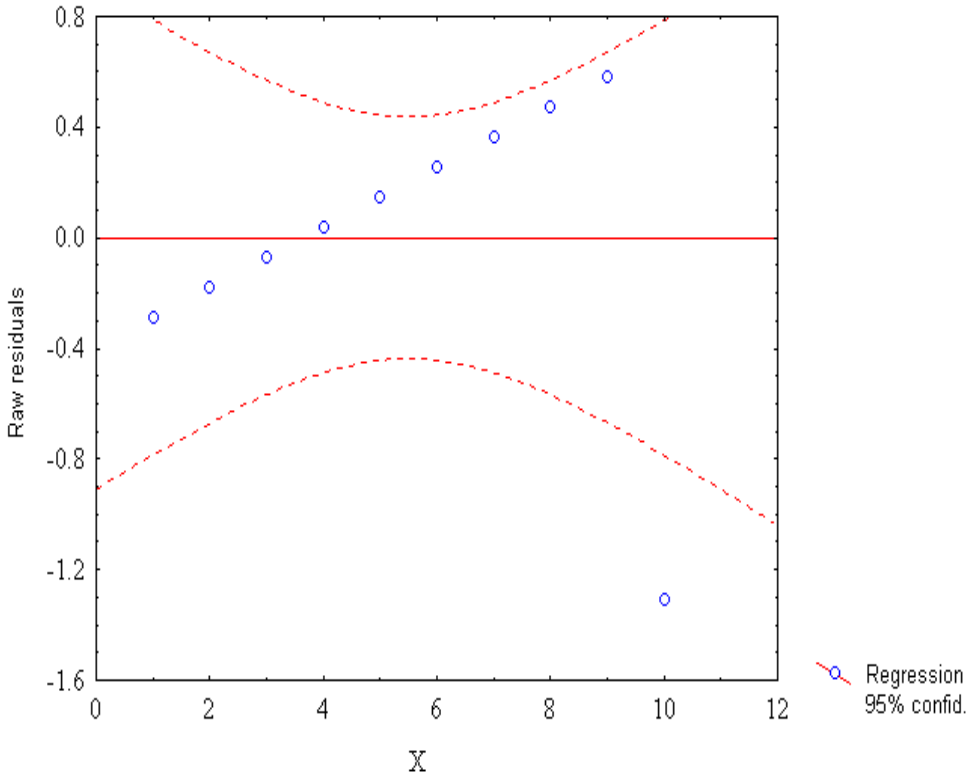


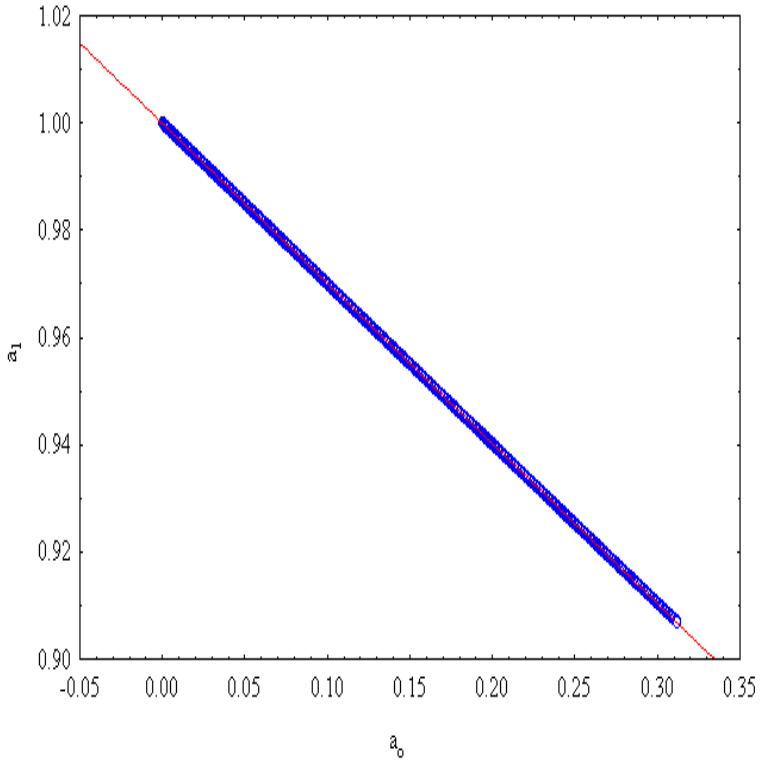
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)

X	Y
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	**8**



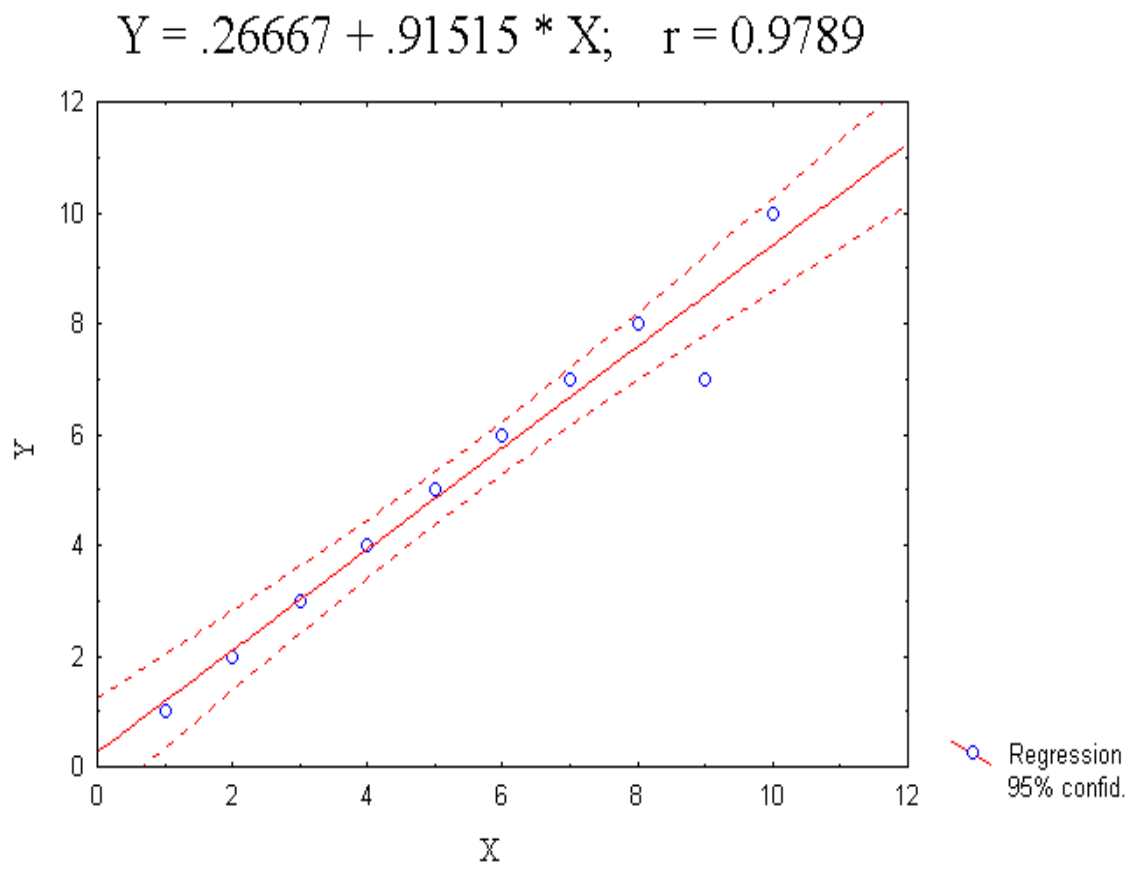
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)



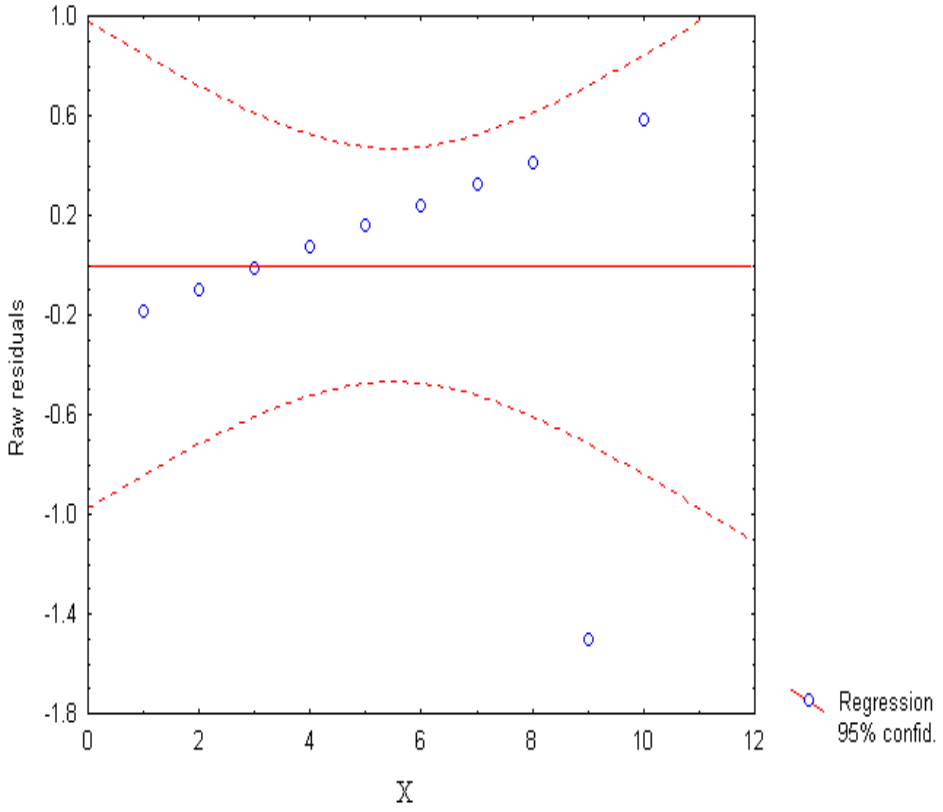


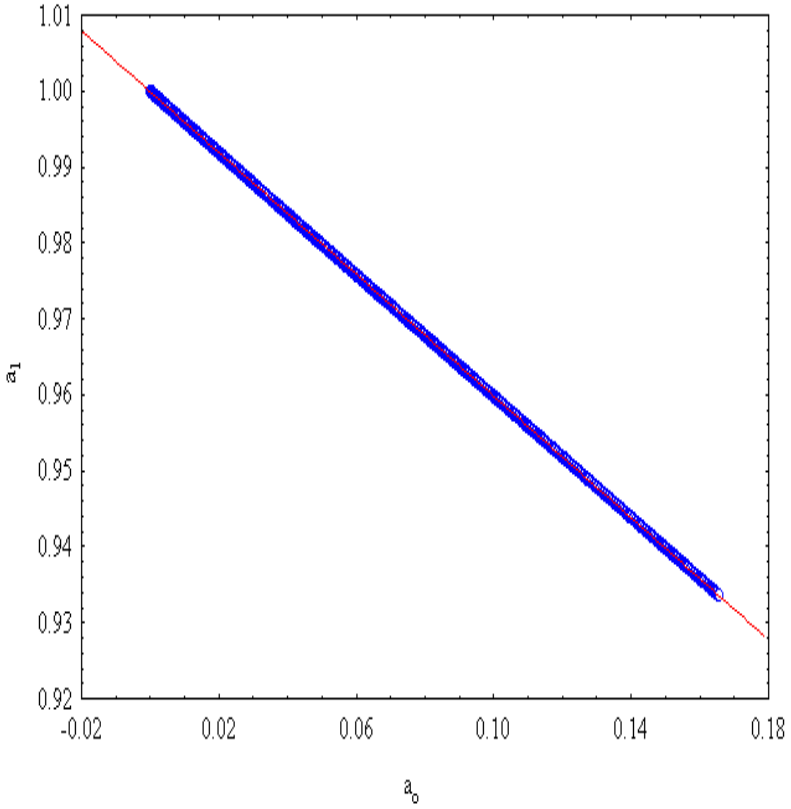
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)

X	Y
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	**7**
10	10



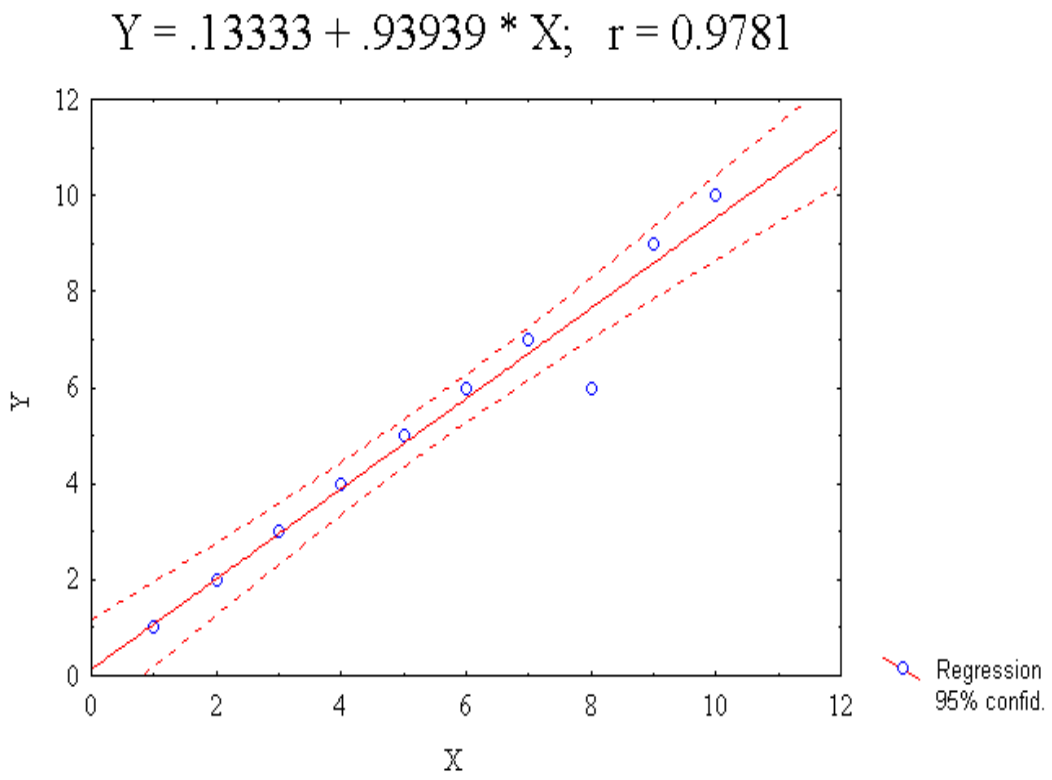
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)



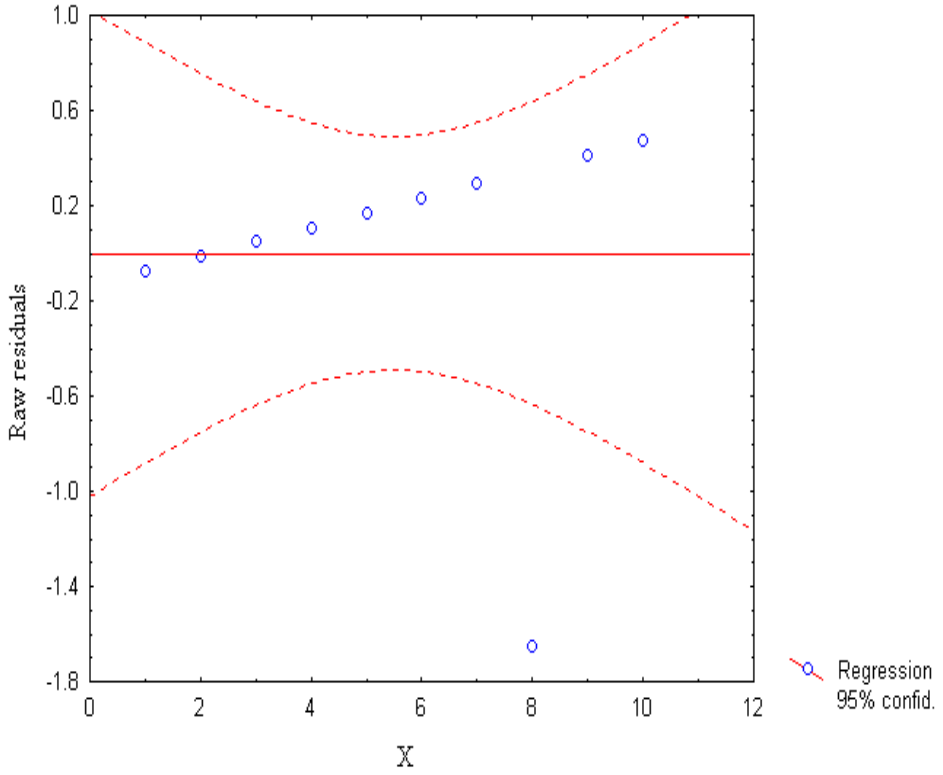


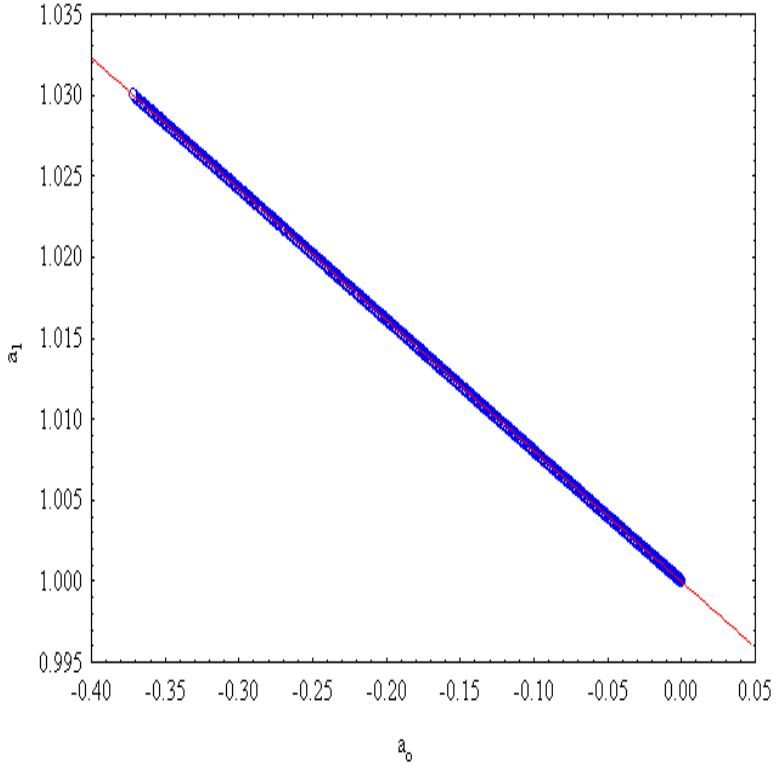
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)

X	Y
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	**6**
9	9
10	10



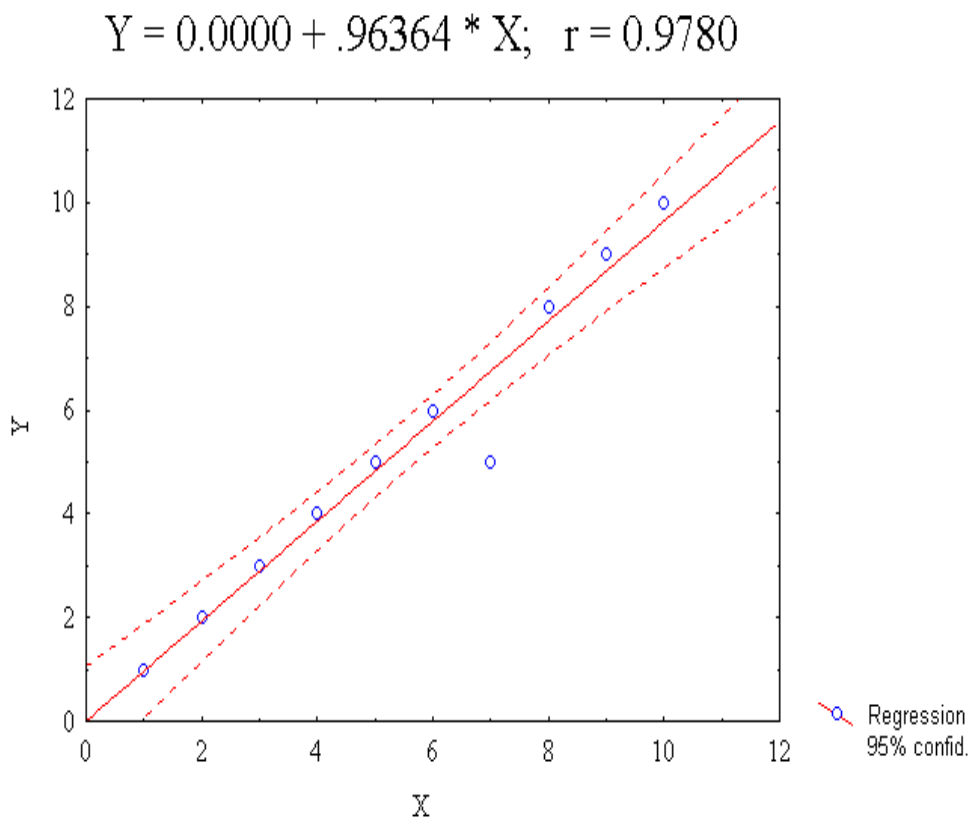
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)



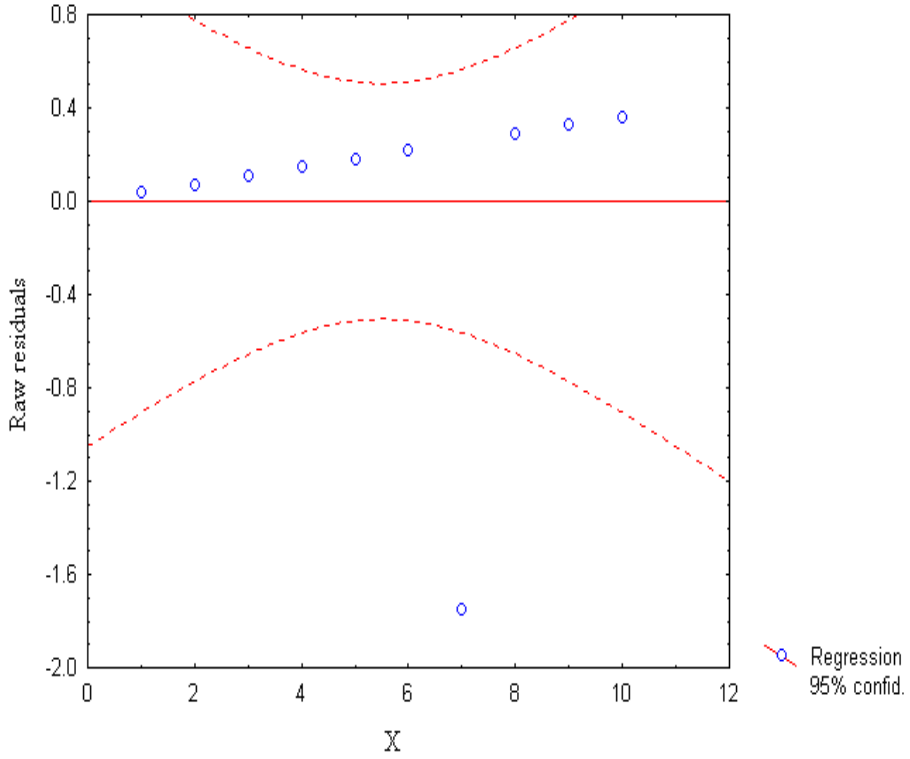


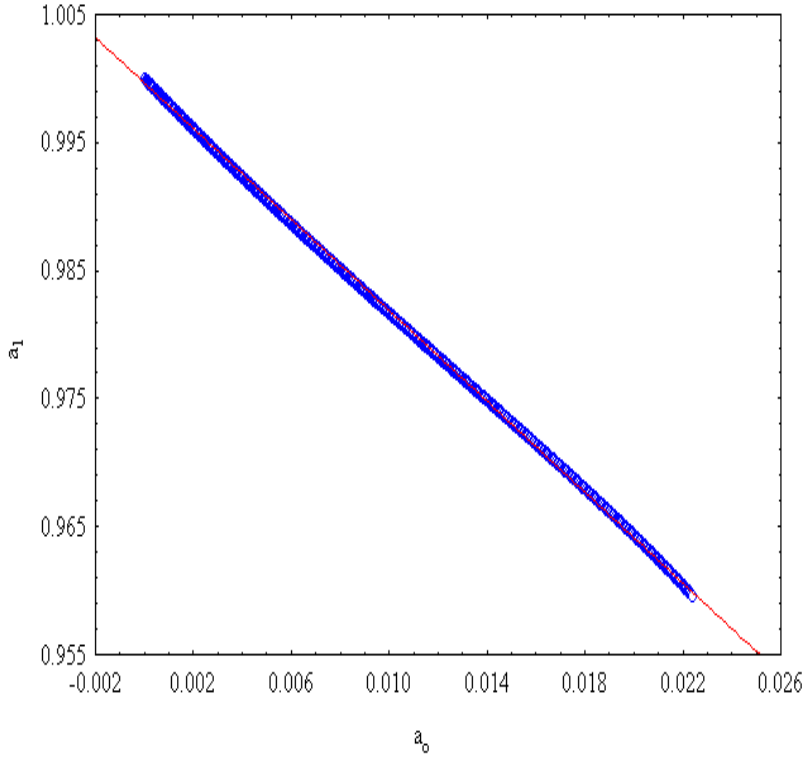
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)

X	Y
1	1
2	2
3	3
4	4
5	5
6	6
7	**5**
8	8
9	9
10	10



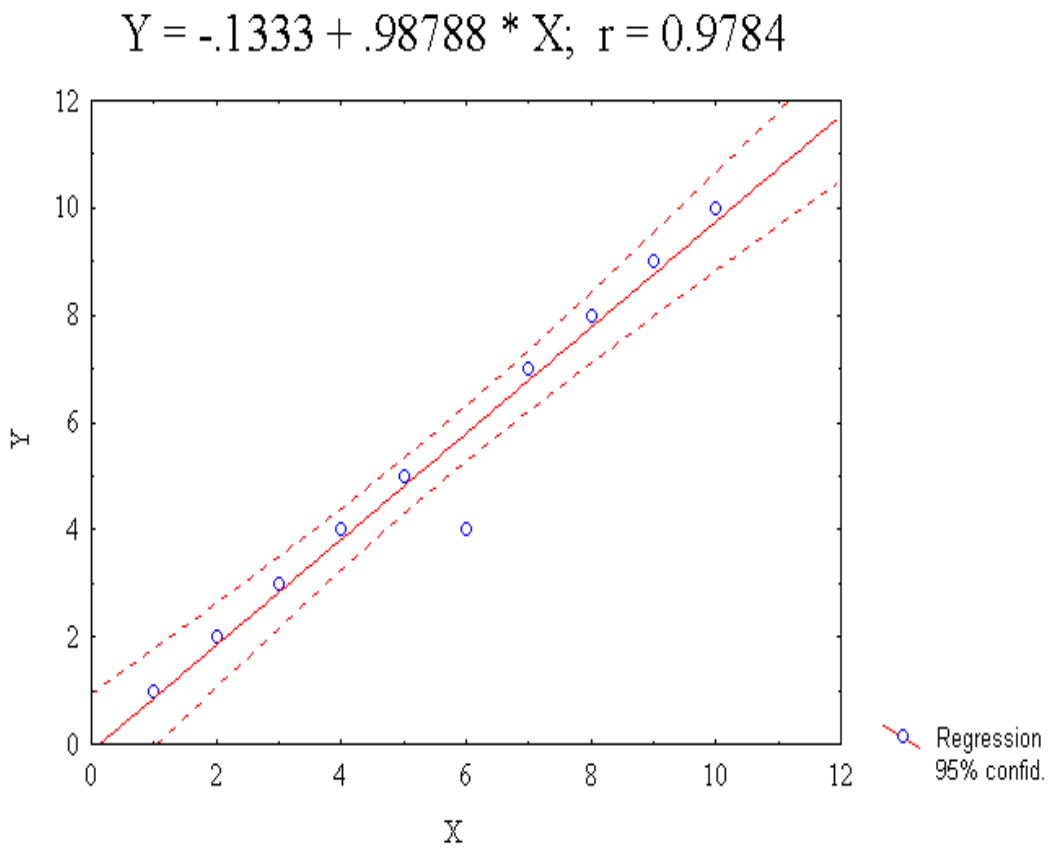
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)



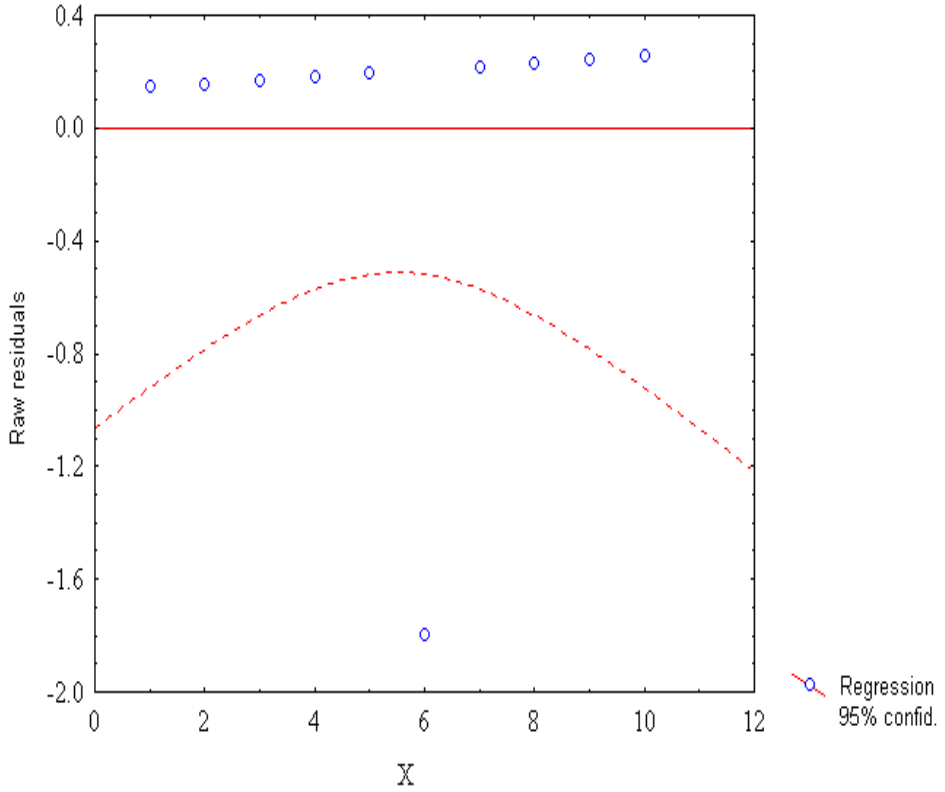


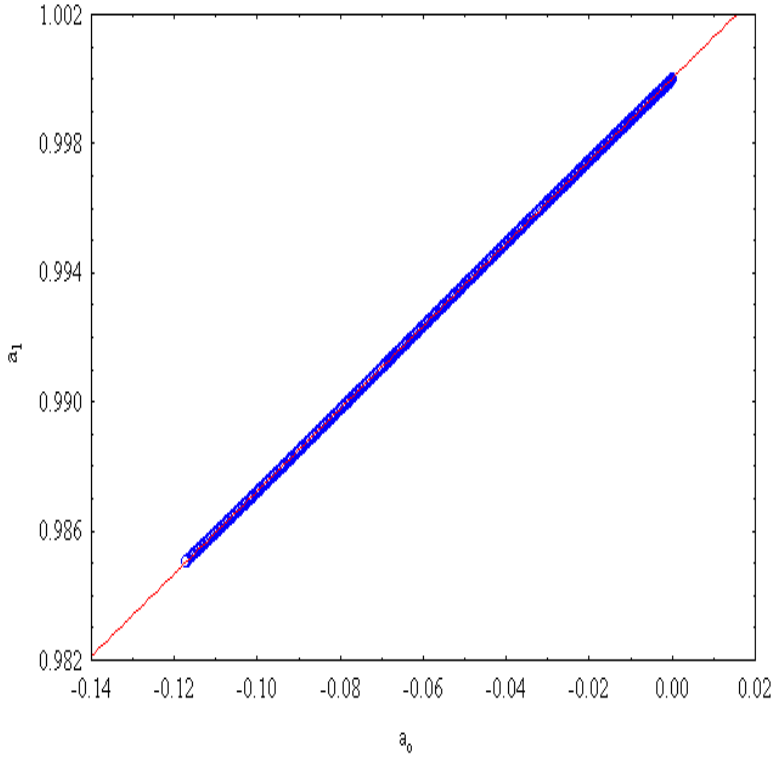
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)

X	Y
1	1
2	2
3	3
4	4
5	5
6	**4**
7	7
8	8
9	9
10	10



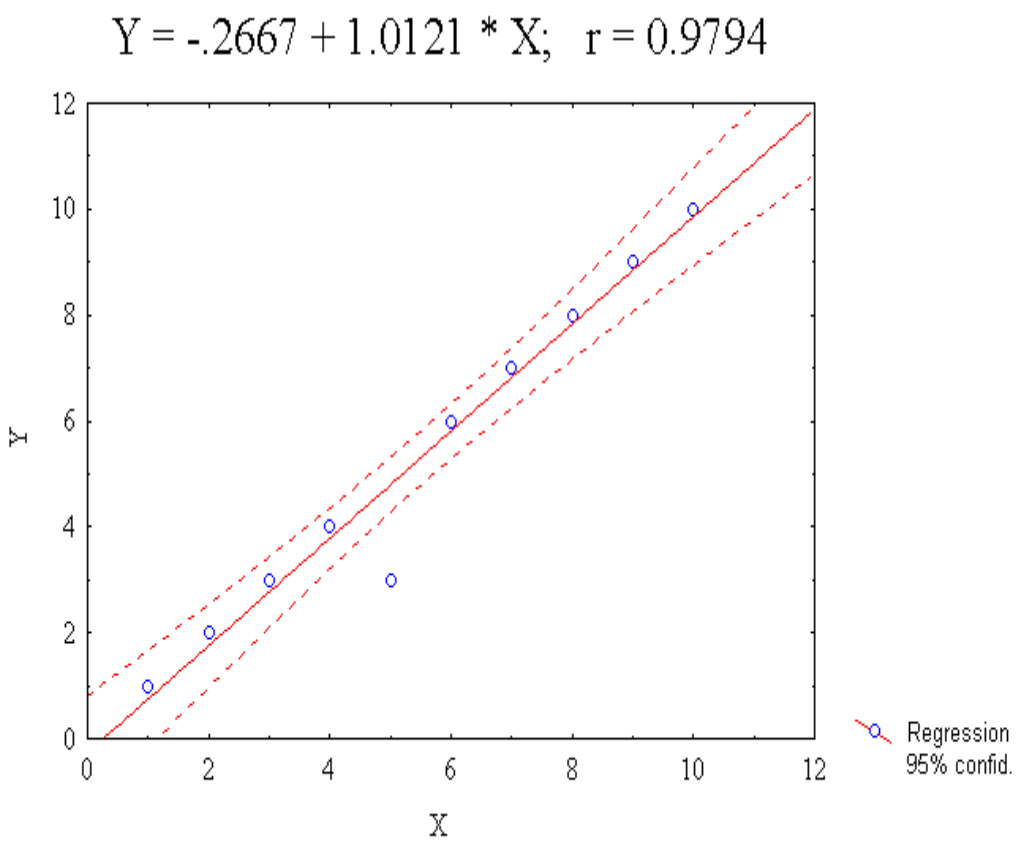
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)



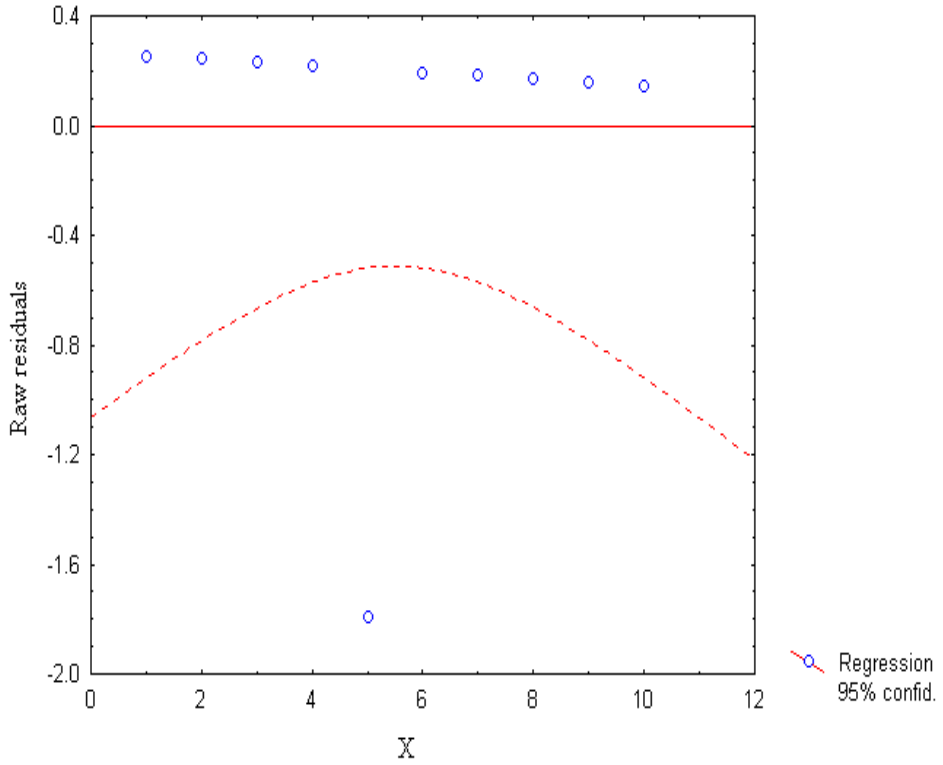


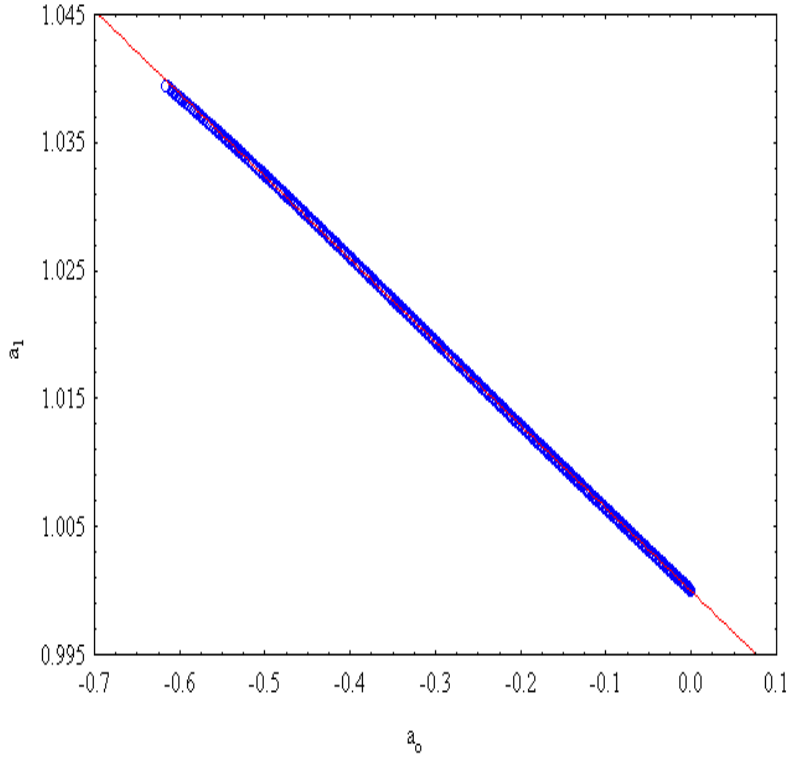
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)

X	Y
1	1
2	2
3	3
4	4
5	**3**
6	6
7	7
8	8
9	9
10	10



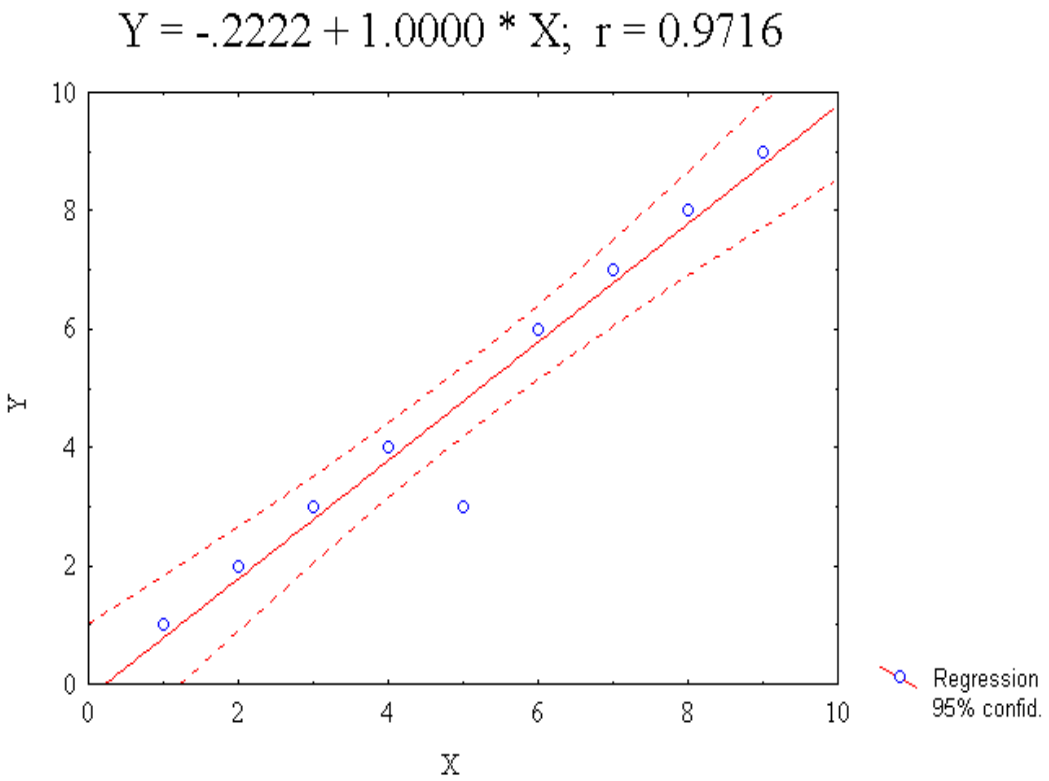
Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)



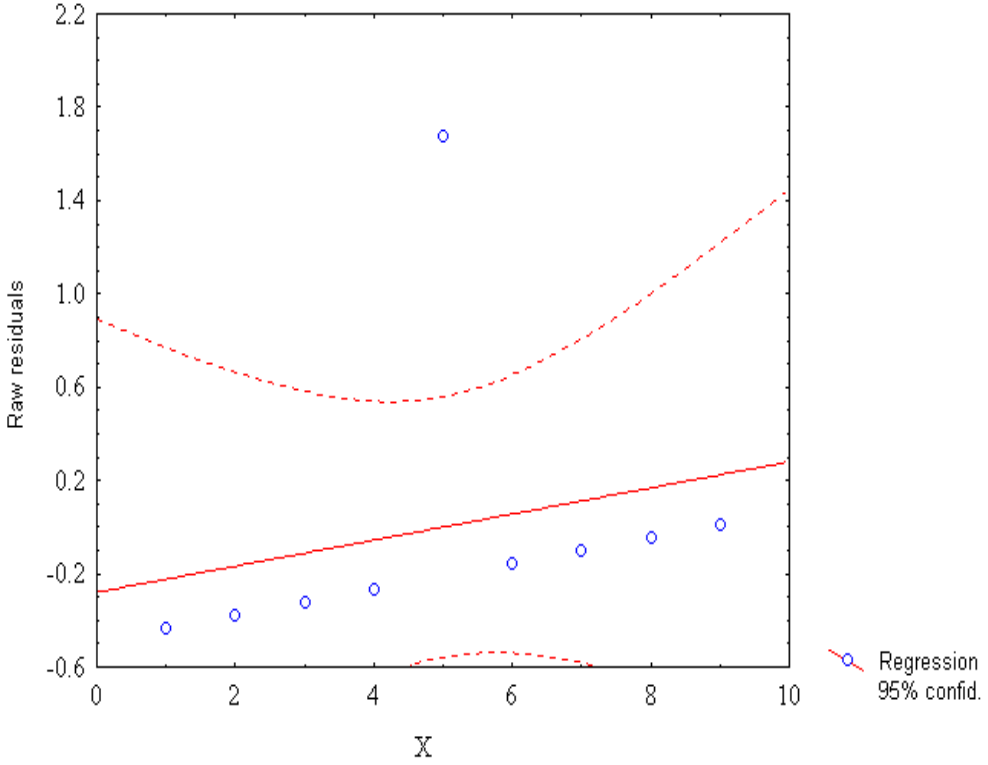


Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)

X	Y
1	1
2	2
3	3
4	**2**
5	3
6	6
7	7
8	8
9	9
10	10



Fuzzy Pattern Recognition of Outliers in Comparing to Residuals Plot (1 Outlier?)



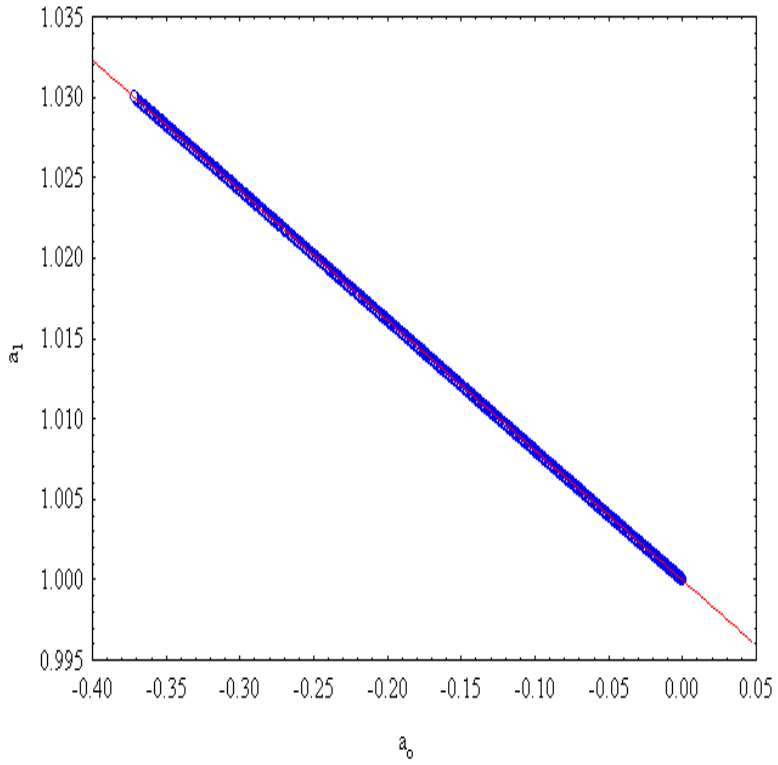


Table of Correlation Between the Regression Parameters (1 outlier?)

a_o	a_1	r
0.400	0.891	0.9806
0.267	0.915	0.9789
0.133	0.939	0.9781
0.000	0.964	0.9780
-0.133	0.988	0.9784
-0.267	1.012	0.9794

	a_o	a_1	r
a_o	1.0		
a_1	**-1.0**	**1.0**	
r	0.4	-0.4	1.0

Fuzzy Clustering

- In this example, we will analyze a set of 32 *in vivo (1-6)* and *in vitro (7-32)* toxicity test systems: human (3,4,21-32), rat (1,5-15), mouse (2, 16-20)

- we used the toxicity data (log 1/C) for the first 10 MEIC (Multicentre Evaluation of *In vitro* Cytotoxicity) chemicals: paracetamol (**1**), acetylsalicylic acid (**2**); ferrous sulfate (**3**), diazepam (**4**), amitriptyline (**5**), digoxin (**6**), ethylene glycol (**7**), methanol (**8**), ethanol (**9**) and isopropanol (**10**).

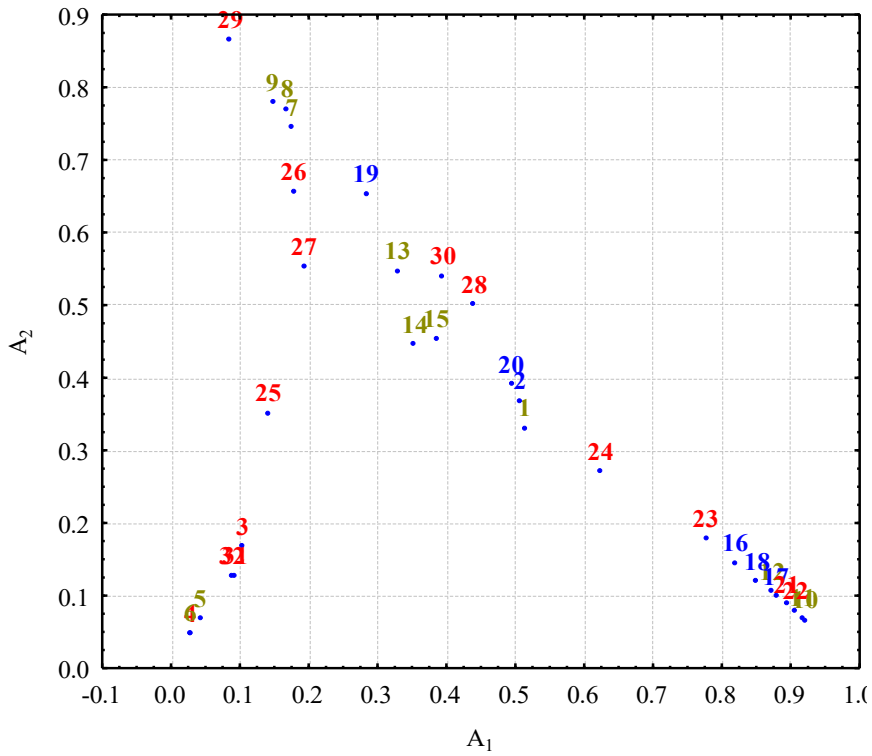
The Hard Partition Corresponding to the Fuzzy Partition

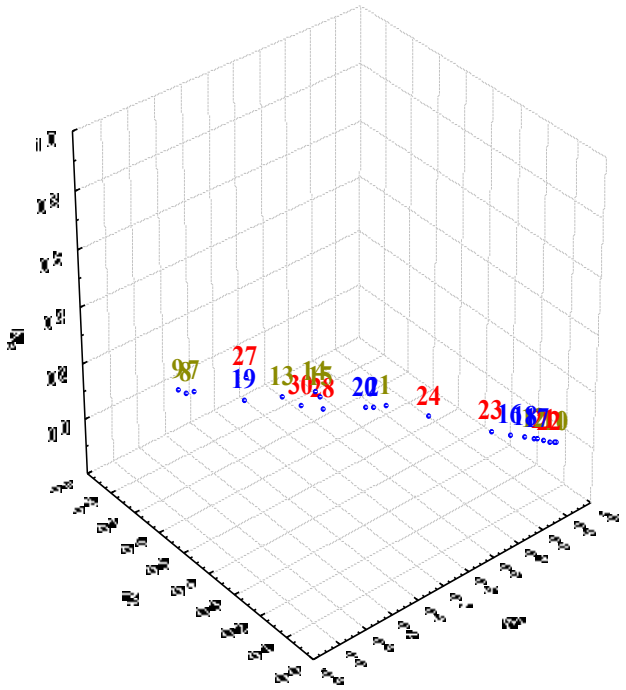
Fuzzy class	Toxicity test systems
A_{11}	7 8 9 13 14 15 19 28 29 30;
A_{121}	1 2 10 11 12 21 22 23;
A_{122}	16 17 18 20 24;
A_{21}	25 26 27;
A_{2211}	3 5;
A_{2212}	4 6;
A_{2221}	31;
A_{2222}	32;

The Cross-Classification of the Toxicological Test Systems (a) and the 10 MEIC Chemicals (b)

Fuzzy class	Toxicity test system (a) and chemicals associated (b)
A_{11}	(a) 7, 8, 9, 10, 13, 14, 15, 19, 28, 29, 30; (b) 1, 2, 3;
A_{121}	(a) 1, 2, 10, 11, 12, 21, 22, 23; (b) 4;
A_{122}	(a) 16, 17, 18, 20, 24; (b) 5, 6;
A_{21}	(a) 25, 26, 27; (b) 7, 10;
A_{221}	(a) 3, 4, 5, 6; (b) 8;
A_{222}	(a) 31, 32; (b) 9;

Membership Degree Plot A_1 vs A_2 and A_1 vs A_2 vs A_3





Principal Component Analysis

Principal component analysis (PCA) is a favorite tool in chemometrics for data compression and information extraction. PCA finds linear combinations of the original measurement variables that describe the significant variations in the data. However, it is well-known that PCA, as with any other multivariate statistical method, is sensitive to outliers, missing data, and poor linear correlation between variables due to poorly distributed variables. As a result data transformations have a large impact upon PCA. In this regard one of the most powerful approach to improve PCA appears to be the fuzzification of the matrix data, thus diminishing the influence of the outliers. Hier, we discuss and apply a robust fuzzy PCA algorithm (FPCA).

Data Set 1

The data collection was performed in the northern part of Romanian Carpathians Mountains : the western part of Bistriţa Mountains (**b**), the south-western part of Maramureş Mountains (**m**) and the north-western part of Igniş-Oaş Mountains (**i**), according to standardized methods for sampling, sample preparation and analysis. Thirteen different soil ion concentration were checked: **lead, copper, manganese, zinc, nickel, cobalt, chromium, cadmium, calcium, magnesium, potassium, iron** and **aluminum**

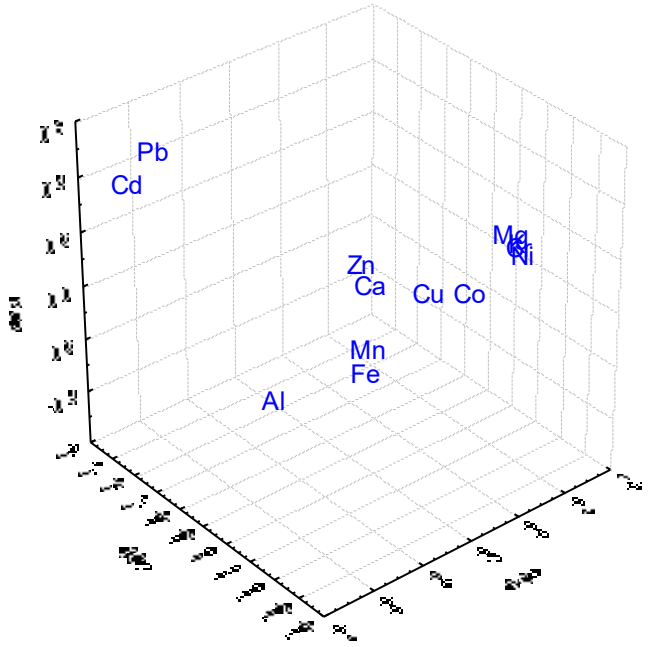
Eigenvalue and Proportion Considering the First Five Principal Components for PCA and FPCA

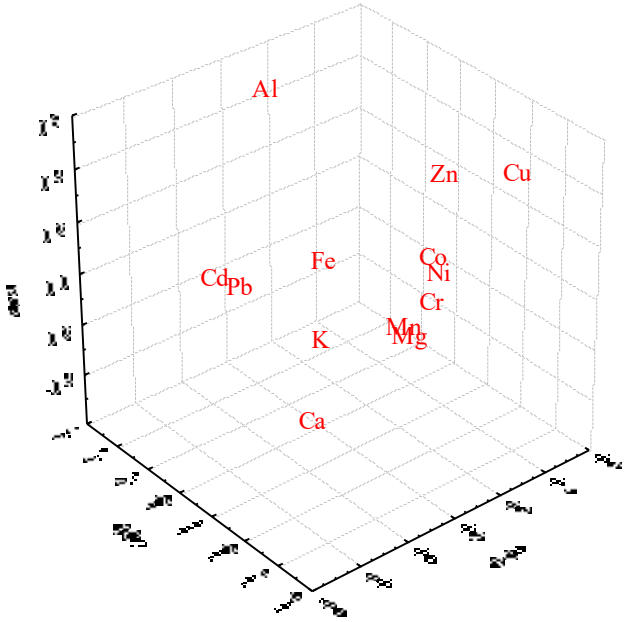
PCs	PCA			FPCA-1			FPCA-o		
	Eigen-value	Prop. %	Cum. Prop. %	Eigen-value	Prop. %	Cum. Prop. %	Eigen-value	Prop. %	Cum. Prop. %
1	5.639	**43.37**	**43.37**	3.161	**48.15**	**48.15**	3.161	**62.78**	**62.78**
2	1.826	**14.04**	**57.42**	0.982	**14.96**	**63.11**	0.724	**14.38**	**77.14**
3	1.403	**10.79**	**68.22**	0.703	**10.71**	**73.82**	0.417	**8.28**	**85.44**
4	1.308	**10.06**	**78.28**	0.554	**8.44**	**82.26**	0.208	**4.77**	**89.57**
5	0.801	**6.16**	**84.44**	0.299	**4.56**	**86.82**	0.240	**4.13**	**94.34**

Eigenvectors Corresponding to the First Four Principal Components for PCA and FPCA

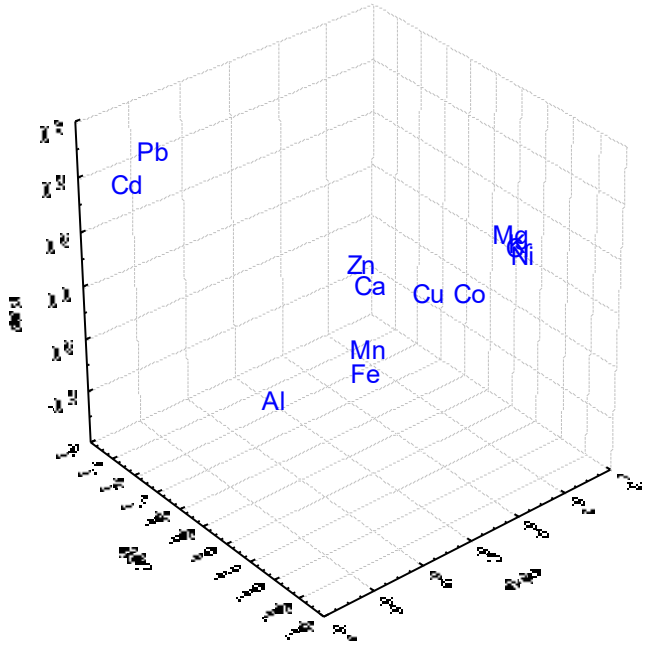
	PCA				FPCA-1				FPCA-o			
	PC1	PC2	PC3	PC4	FPC1	FPC2	FPC3	FPC4	FPC1	FPC2	FPC3	FPC4
Pb	-0.065	**0.451**	**0.539**	-0.165	-0.019	0.045	0.131	**0.403**	-0.019	-0.025	**-0.589**	-0.089
Cu	0.277	0.030	-0.004	**-0.457**	**0.391**	**-0.415**	**0.419**	0.046	**0.391**	**0.341**	-0.086	**-0.416**
Mn	0.265	0.251	**-0.340**	0.206	**0.409**	0.260	**-0.477**	-0.144	**0.409**	-0.205	0.127	**0.481**
Zn	**0.311**	**0.372**	-0.124	-0.119	**0.470**	0.196	0.114	0.186	**0.470**	-0.179	-0.164	-0.081
Ni	**0.402**	-0.105	0.111	-0.046	**0.300**	-0.221	0.035	0.019	**0.299**	0.222	-0.006	-0.090
Co	**0.397**	0.091	-0.139	0.078	**0.404**	0.079	-0.112	-0.086	**0.404**	-0.061	0.090	0.094
Cr	**0.362**	-0.159	0.206	-0.097	0.240	**-0.341**	0.022	0.043	0.240	**0.317**	-0.003	-0.100
Cd	-0.058	**0.585**	**0.345**	0.032	0.013	0.296	0.034	**0.809**	0.013	-0.234	**-0.743**	0.094
Ca	0.175	0.066	0.088	**0.609**	0.127	0.041	**-0.519**	0.058	0.127	0.058	-0.041	**0.607**
Mg	**0.380**	-0.095	0.201	0.136	0.255	-0.183	-0.190	0.124	0.255	0.230	-0.059	0.148
K	**0.311**	-0.245	**0.309**	0.072	0.049	-0.228	-0.007	0.043	0.049	0.219	-0.016	-0.044
Fe	0.101	-0.063	-0.095	**-0.541**	0.111	-0.072	0.170	-0.038	0.111	0.012	0.014	-0.177
Al	0.121	**0.359**	**-0.481**	-0.027	0.226	**0.607**	**0.463**	**-0.302**	0.226	**-0.704**	0.192	**-0.349**

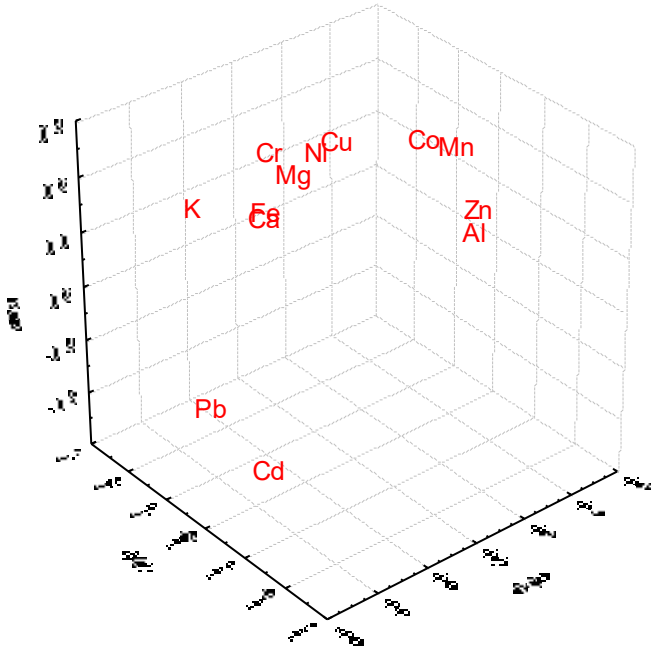
Loading Plot PC1-PC2-PC3 (PCA and FPCA-1)



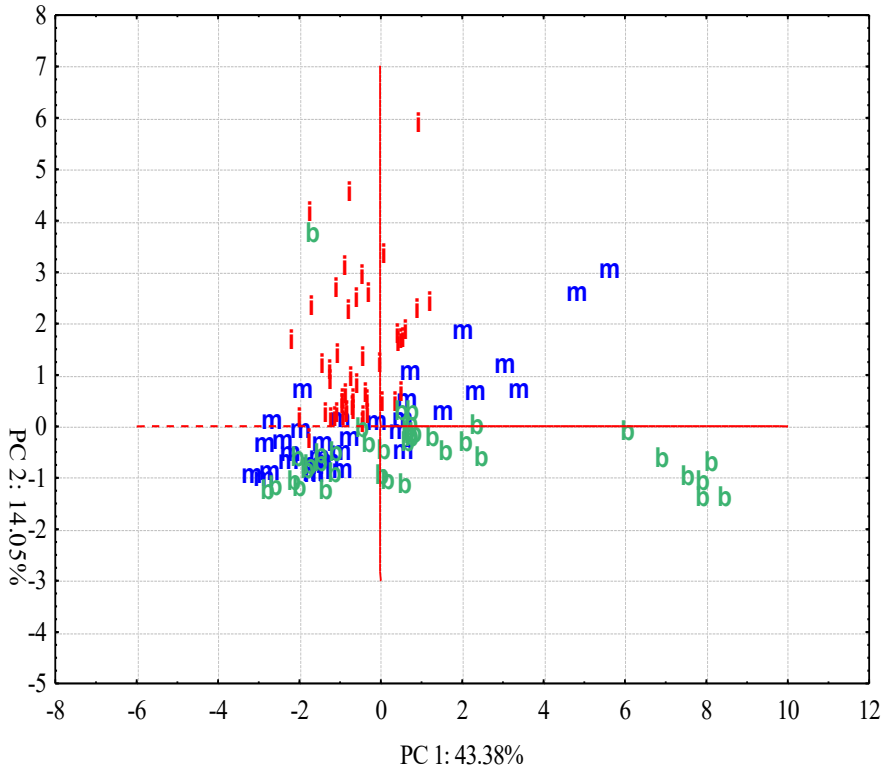


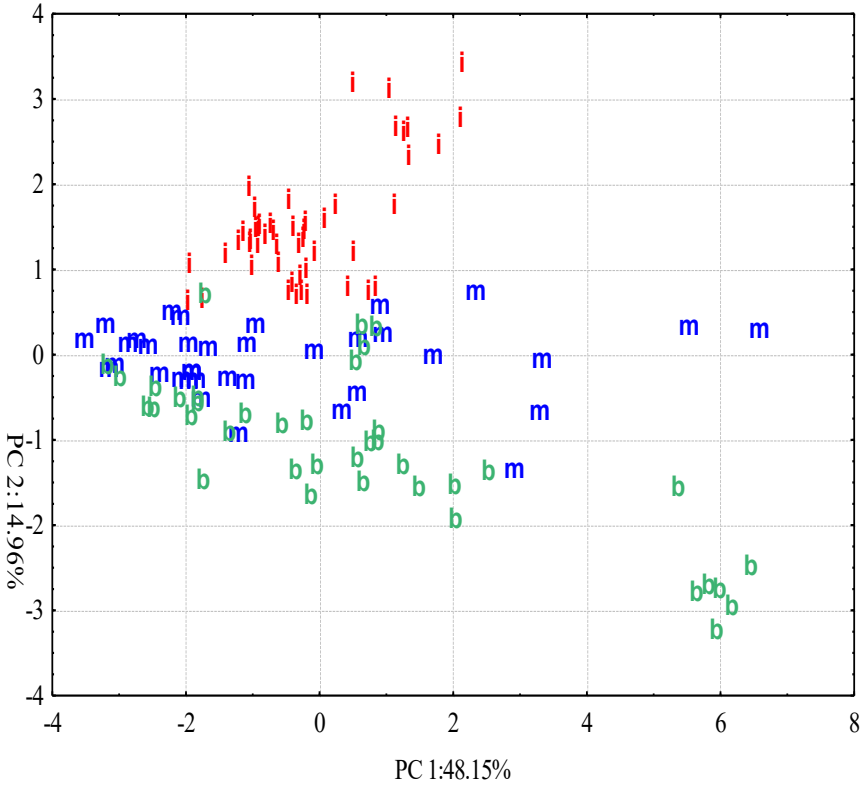
Loading Plot PC1-PC2-PC3 (PCA and FPCA-o)



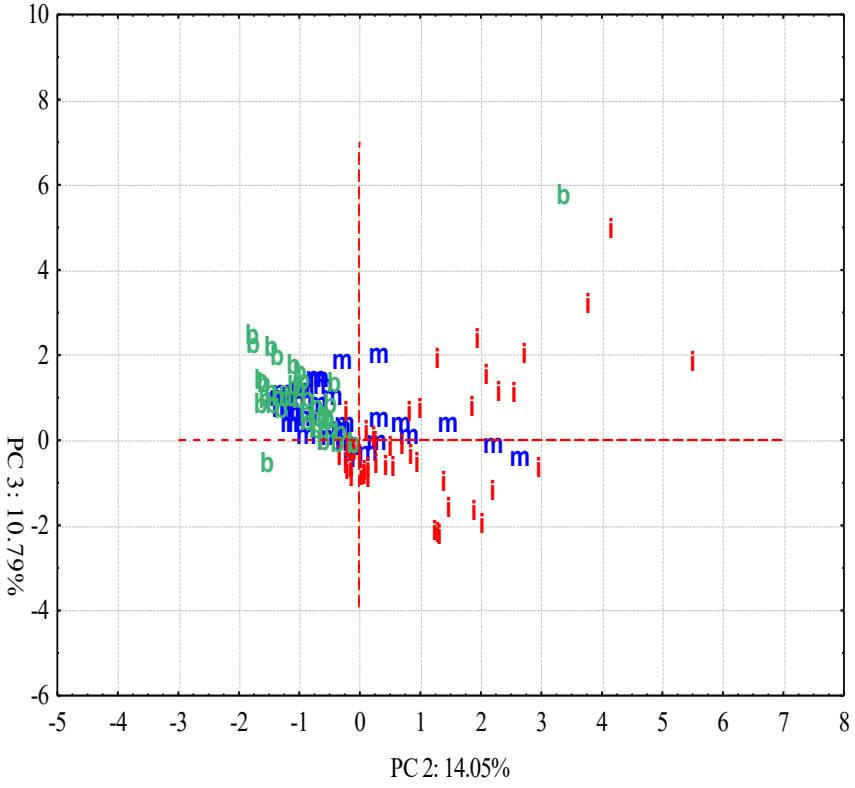


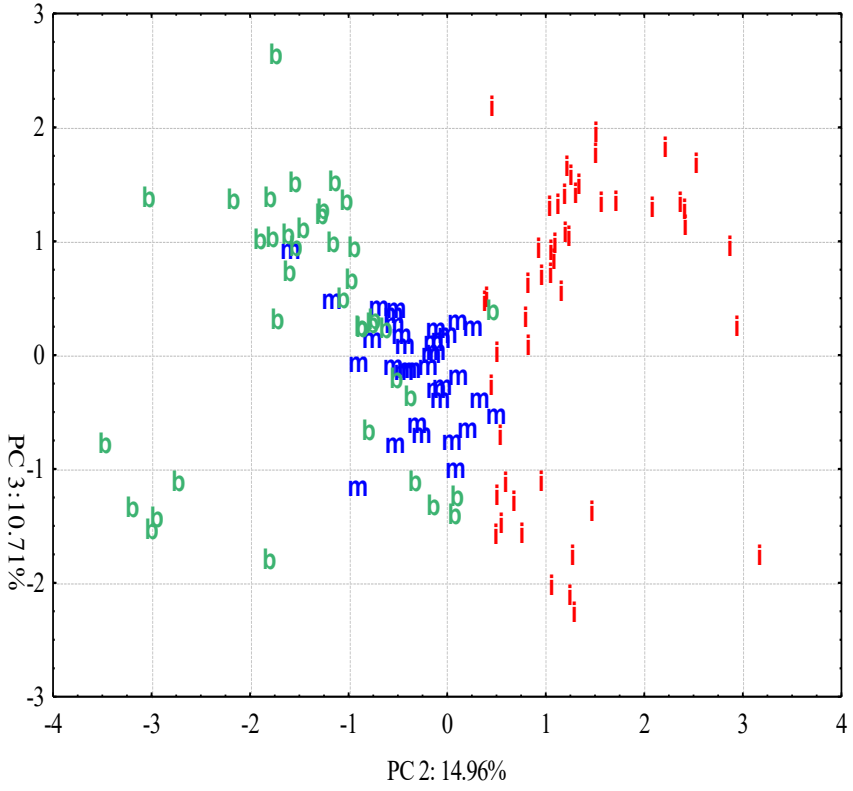
Score Plot PC1-PC2 (PCA and FPCA-1)



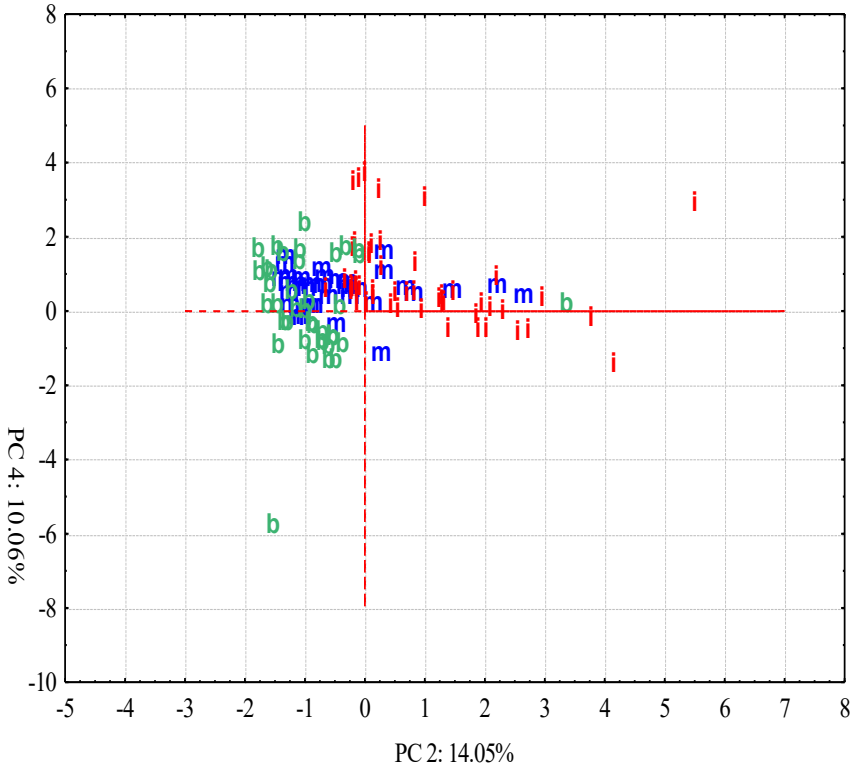


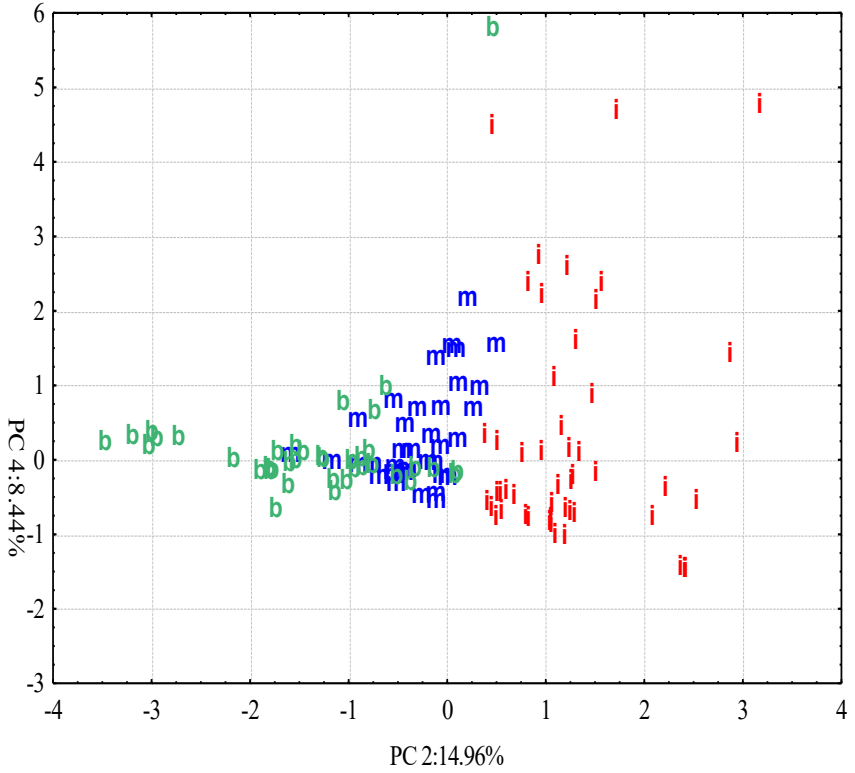
Score Plot PC2-PC3 (PCA and FPCA-1)



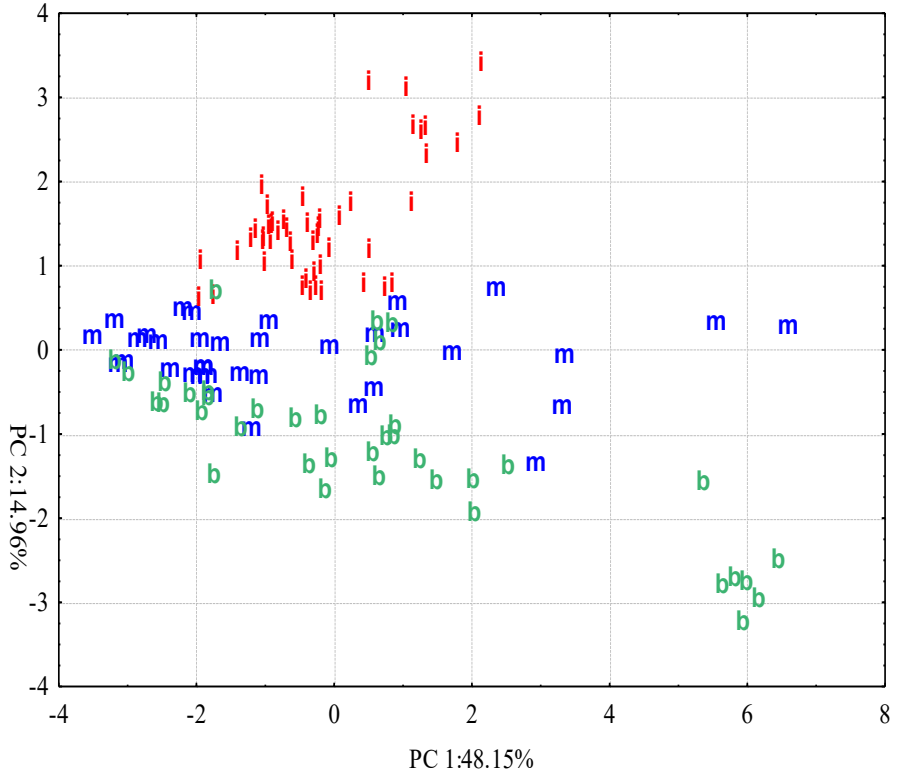


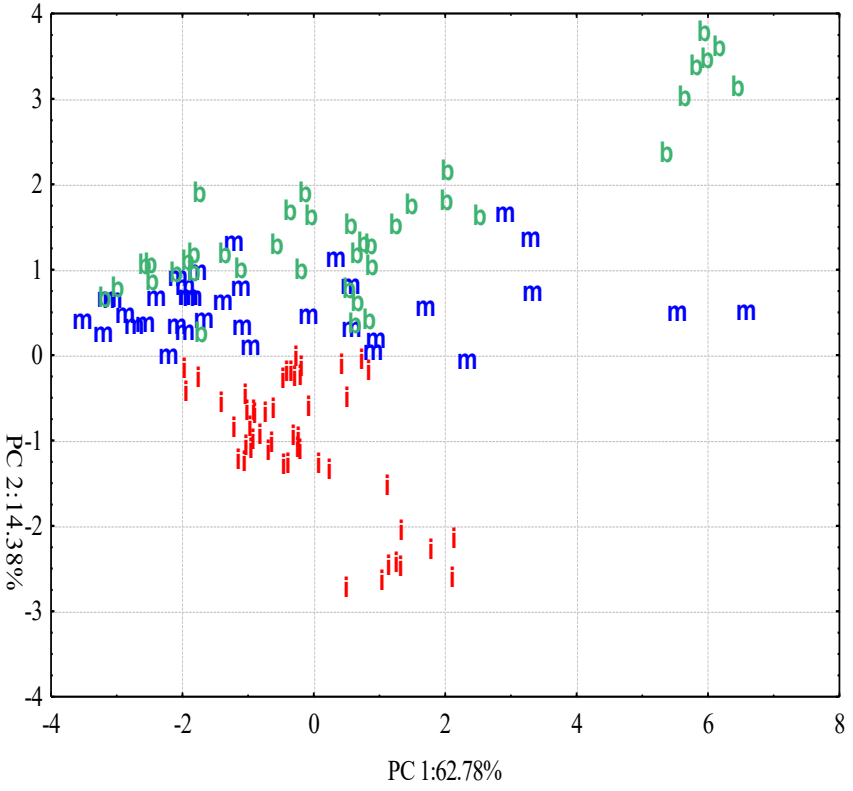
Score Plot PC2-PC4 (PCA and FPCA-1)





Score Plot PC1-PC2 (FPCA-1 and FPCA-o)





Concluding Remark

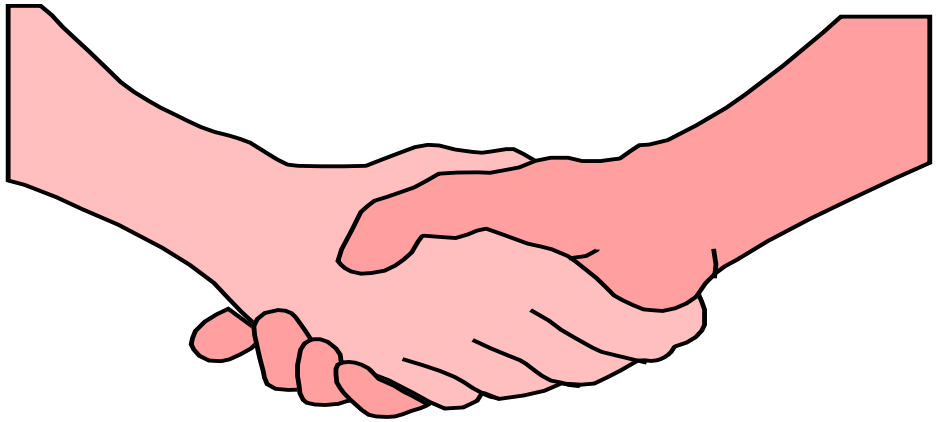
- **"Are the Concepts of Chemistry all fuzzy?"**

(*The title of the Conference organized by Rouvray and Kirby, 1995*)

- **If yes, then Fuzzy Soft Computing** could be one of the best solution for solving problems in chemistry and related fields!?

Soft Computing and the Future

- **Soft computing** and **conventional methods** should be used in combination



The Bright Future of Chemometrics

The responsibility for change …
lies within us. We must begin with ourselves, teaching ourselves not to close our minds prematurely to the novel, the surprising, the seemingly radical.

Alvin Toeffler



Thank you for your attention!

